

The World of Thermoregulation

huber

Ministat 125, 230, 240 - efficient thermal regulation



**ministat 125
ministat 230
ministat 240**

Operating Instructions

Version 1.0/04

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EC Declaration of Conformity

We declare that the design and model of the thermostat described in the following and the version put into circulation by complies with all the relevant and applicable safety and health requirements laid down in the corresponding EC directive.

If the thermostat is modified without the modification being agreed upon by the manufacturer, this declaration will become void.

Model	Comp. Control Thermostat Ministat 125	Order no. 740.000X
	Comp. Control Thermostat Ministat 230	Order no. 741.000X
	Comp. Control Thermostat Ministat 240	Order no. 742.000X
	Series 03/04	
Identification		
EC Directives	EC Low Voltage Directive 73/23/EEC 93/68/EEC amendment	
	EC Electromagnetic Compatibility Directive 89/336/EEC 92/31/EEC, 93/68/EEC, 98/13/EEC amendment	
Harmonized Standards	EN 61010-1 EN 61010-2-010 EN 61326	
National Standards and Technical Specifications	DIN 12876-1 DIN 12876-2 DIN 12876-3	
Manufacturer:	Peter Huber Kältemaschinenbau GmbH Werner-von-Siemens-Straße 1, D-77656 Offenburg	
	01.11.2003, CEO Daniel Huber	



Symbols



Safety

Warning! A potentially hazardous situation. Identifies hazards sufficient to cause death or severe injuries if the safety instructions are disregarded.

Caution! A potentially hazardous situation. Identifies hazards sufficient to cause light injuries if the safety instructions are disregarded.



Definitions from our Huber Glossary and fundamental technical knowledge.



Stepwise instructions for operating the device and the controller.



Entries at the Polystat cc controller.



Device messages.



Practice.



Service

This is where you get help: the Huber Hotline.



EXTRA

Additional information.

Preface

Dear Customer,

Congratulations! Units and devices manufactured by Peter Huber Kältemaschinenbau GmbH are always a good choice. Thank you very much for your trust.

To meet your demands as a user, we have revolutionized the user interface of our thermostats and implemented a uniform method of operating almost the entire HUBER product range. Many devices, ranging from small immersion thermostats to large Unichillers are operated via a single controller generation: Polystat Control. The controller for Ministats 125, 230 and 240 **Ministat Control** was created in the course of this further development. It offers all the functions and convenience of the Polystat Control and can be operated just as easily, which brings us back to the concept of uniform operation.

On the type label on the rear of your device you will find important information such as:

ministat 240	-40°... +200°C	SNr.: 55655/03
[Device name]	[Temp.range]	[Serial number]

Free choice in operation:

All the Ministats can be operated with three different controller versions:

Ministat Control cc1, the simple one

Ministat Control cc2, the convenient one

Ministat Control cc3, the one with dialog capabilities

Your controller version is identified by the label above the display. For details on your type of controller, please refer to Chapter 2.2

Please consider only those instructions in the present documentation that apply to your device type and controller version.

Preface



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Technical Data Sheet

List of Spare Parts



1. Safety

1.1. Intended Use

General Safety Instructions



The thermostat is designed for industrial applications.

The thermostat is used for direct and indirect thermoregulation, i.e. for heating or cooling external substances through suitable thermal fluids. It must be operated strictly in compliance with the operating instructions.

The thermostat must not be modified by the plant operator or any operating personnel.

The thermostat must not be used for purposes other than thermoregulation in compliance with the operating instructions. Unintended use or use not in compliance with the operating instructions may lead to severe personal injury or property damage.



Your device has been designed and constructed according to the state of the art and in compliance with the generally accepted safety rules. Nonetheless, your device may constitute an imminent or unexpected hazard. For this reason, your device has been equipped with safety devices. Deactivating these safety devices bears high risks and invalidates the warranty.



Use the device only if it is in good order and condition.

Shut down the device immediately in the case of malfunctions or failures.

Only qualified personnel is permitted to perform repairs.

Do not bypass, bridge, dismount or deactivate any safety devices.

The manufacturer assumes no liability for damage due to technical modifications, improper handling or use of the device disregarding the operating instructions.

The manufacturer assumes no liability for damage due to technical modifications, improper handling or use of the device disregarding the operating instructions.



Warning! Risk of injuries!

While operating at high temperatures, the bath lid and the housing could become very hot.

Only touch the housing and the lid by the grips otherwise there is a *risk of burns*!

Never lift the bath's lid during operation at high temperature:

- Risk of *scald /burn* through thermal fluid overflow.

Depending on the type of thermal fluid used:

- Risk of *caustic vapours* causing injuries to the respiratory tract and/or skin!

(For further information about the chosen thermal fluid please read the material safety data sheet delivered with it.)

This warning is only applicable for units with this  warning sign.



Important: transport damage!

When unpacking the device, inspect it for transport damage.

Please revert to the haulage contractor or shipping agent for settlement of claims.

Commission a damaged device only after the damage has been repaired or you have ascertained the full effects of damage and the insurance agent/haulage contractor/shipping agent has given their permission.



1. Safety

1.2. Intended Use

General Safety Instructions



Duties of the Plant Operator:

-  The operating instructions must be kept readily available in the immediate vicinity of the thermostat.
-  Only sufficiently qualified operating personnel are permitted to use the thermostat.
-  The operating personnel must be trained in handling and using the thermostat.
-  Verify that operating personnel have read and understood the operating instructions.
-  Precisely define the fields of responsibility of the operating personnel.
-  Provide protective clothing for the operating personnel.



Requirements to be Met by the Operating Personnel

-  Only personnel assigned and trained by the plant operator may handle and operate the thermostat.
-  The minimum age for operating personnel is 16 years. Within the workspace, the device operator is responsible for third parties.
-  The device operator must be sufficiently qualified



Duties of the Operatoring Personnel:

-  The operating personnel must read the operating instructions thoroughly before handling or using the thermostat.
-  The operating personnel must heed all the safety instructions.
-  The operating personnel must wear protective clothing when handling or using the thermostat.



Workspace

The workspace is defined to be at the control panel in front of the thermostat. The workspace is further defined by the peripheral equipment connected by the customer. The customer is responsible for taking suitable safety measures.



Safety Devices

-  Over-temperature protection device
-  Low liquid level protection
-  Mains failure automatic
-  Alarm functions



Emergency Plan – Switch off the Power Supply!

Hazardous emission of fluid/vapor from the thermostat or connected pipes/ hoses (very hot, very cold, hazardous chemical compositions) and or fire/ explosion/implosion:

-  Strictly heed the safety instructions of the plant operator relating to the risk of injury and danger to life as well as to the limitation of damage.
-  Observe the instructions included in the safety data sheet of the respective thermal fluid!

2

Device Description





2. Device Description

2.1. Structure

Compatible Control Thermostat ministat 125

Compatible Control Thermostat ministat 230

Compatible Control Thermostat ministat 240

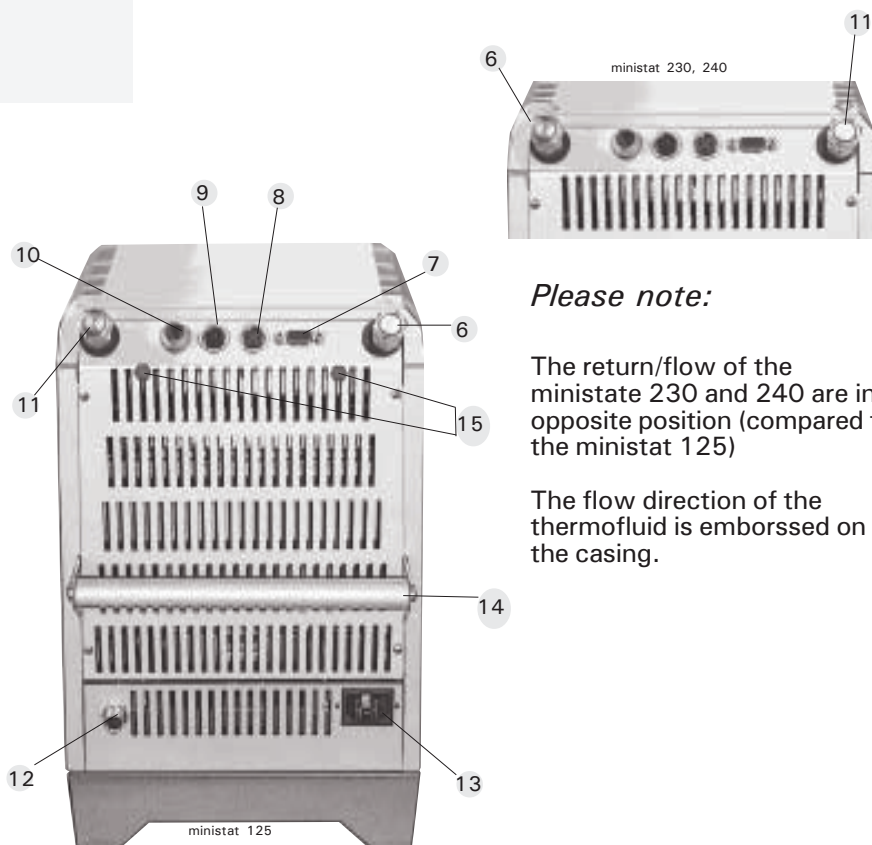
Working temperature range of the ministat 125: -25...150°C.
Working temperature range of the ministat 230: -33...200°C.
Working temperature range of the ministat 240: -40...200°C.

- 1 Detachable, exchangeable Ministat Control
- 2 Bath lid
- 3 Draining connection

View of RH front side



View of rear side



Please note:

The return/flow of the ministate 230 and 240 are in opposite position (compared to the ministat 125)

The flow direction of the thermofluid is embossed on the casing.

	NAMUR designation
6 Return	
7* RS232/485	SERIAL
8* ECS (External control signal)	STANDBY
9* AIF (Analog Interface)	REG + E-PROG
10* POKO (Potenzialfreier Kontakt)	ALARM
11 Flow	
12* CAN connection	
13 Power supply connection	
14 Handle	
15 Attachment screw (for the Ministat Control)	

* The interface can be fitted optionally



2. Device Description

2.2. Controller

Ministat Control cc1
Ministat Control cc2
Ministat Control cc3

Display, control panel

R1 Digital status display

R2 Prompt for a temperature set-point, input confirmation (data transfer)

R3 Call of the user menu for convenient handling

R4 On/Off key to start/stop thermoregulation

R5 Activity indicator LEDs

R6 over-temperature protection device

R7 Encoder/ rotate:
Entry of Program numbers, step numbers, parameters (e.g. temperature set-point)

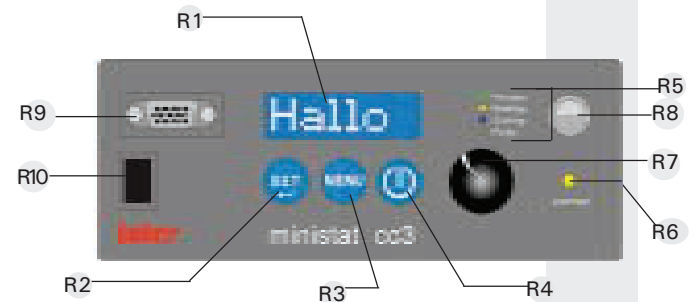
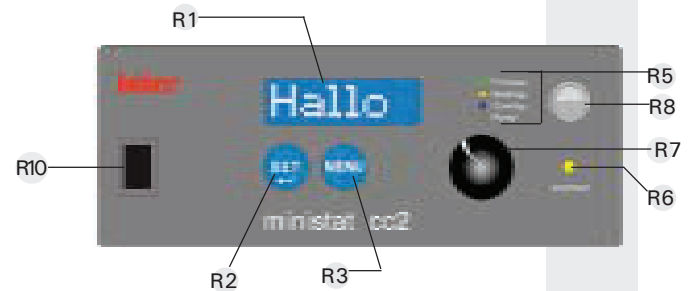
Encoder/ press:
Input confirmation (data transfer)

Connections

R8 Pt100 sensor socket

R9 RS 232/485 interface

R10 Power switch



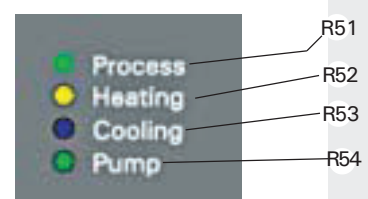
Activity indicator LEDs

R51 Process temperature control active (green LED), only with cc2 and cc3:
Temperature is measured by a process sensor located at the point of control, e.g. in a reactor.

R52 Heating active (yellow LED)

R53 Cooling /compressor active (blue LED)

R54 Pump active (green LED):
Thermal fluid is being pumped through the connected application, e.g. around the reactor jacket.



3

Commissioning





3. Commissioning

3.1. Safety Instructions and Principles

Preparatory Measures for Commissioning

Plan the thermoregulation target and procedure.
Determine the device configuration and system structure.
Select an appropriate thermal fluid.
Selection criteria for thermal fluids: Temperature range of the thermostat, application restrictions building safety regulations, your projected working temperature, viscosity, flash point. Position the thermostat and external devices on a stable and even surface. Ensure that the surface can safely hold the weight of the thermostat and

Positioning

Caution! Potential risk of injury and material damage:

Keep the device upright during transport.
Place the device in an upright and stable position and make sure that it cannot tilt over.
Keep the vicinity of the device clean: Prevent slip and tilting hazards.
Lock the wheels of floor-mounted devices once you have positioned them as desired!

Useful note: Place appropriately large collecting trays under the thermostat and the application.

Location

Place the device on a dense, level and non-slip surface.
Place the device at a distance to walls and the ceiling that permits sufficient air circulation (heat dissipation, supply of fresh air for the thermostat and the workspace). A water-cooled unit requires a minimum of 10 cm and an air-cooled unit needs a minimum of 20 cm.
Ambient temperature max. 30 °C.
Ensure short distances to the supplies.
The workspace of the thermostat must comply with local safety regulations (ArbStättV 20. März 1975 zuletzt geändert BGBl. I 1996)



Please Note:

All the safety instructions are vital and must be considered during the operation of the unit in compliance with the operating instructions.



3. Commissioning

3.1. Principles, Media and Safety Instructions



The operating instructions include additional safety instructions. These are identified through a triangle with an exclamation mark. Thoroughly read and heed the instructions. Non-observance may involve considerable consequences such as device damage, physical damage or personal injury with fatal consequences.

Workspace

The workspace is defined to be at the control panel in front of the thermostat. The workspace is further defined by the peripheral equipment connected by the customer. The customer is responsible for taking suitable safety measures.

Safety Devices

- ^ Overtemperature protection
- ^ Low liquid level protection
- ^ Mains failure automatic
- ^ Alarm functions

Hazardous emission of fluid/vapor from the thermostat or connected pipes/hoses (very hot, very cold, hazardous chemical compositions) and or fire/explosion/implosion:

Strictly heed the safety instructions of the plant operator relating to the risk of injury and danger to life as well as to the limitation of damage.

Observe the instructions included in the safety data sheet of the respective thermofluid!

DIN
12876

Classification according to DIN12876:			
Classification	Thermoregulation fluid	Technical Specifications	Identification ^d
I	non-combustible ^a	Overheating protection	NFL
II	combustible ^b	Adjustable overheating protection	FL
III	combustible ^b	Adjustable overheating protection Additional low-level protection	FL

- a Generally water, other fluids only if they are not combustible in the temperature range of an individual fault.
- b The thermoregulation fluids must have a combustion point of $\pm 65\text{ °C}$, i.e. when using ethyl alcohol, only supervised operation is possible.
- c The overheating protection can be achieved e.g. through an appropriate fill level sensor or appropriate temperature control devices.
- d Optional according to the selection of the manufacturer

Your thermostat is classified as FL / III

We cannot guarantee that our components (individually or combined) are suitable for use in potentially explosive atmospheres.



Please Note: All the safety instructions are vital and must thus be considered on the job in compliance with the present operating instructions..



3. Commissioning

3.1. Principles, Media and Safety Instructions

Not suitable for use as a medical device (e.g. in vitro diagnostic procedures).

Requirements for thermofluids classified as FL:
EN 61010-1: Max. permissible working temperature
25 °C below the flash point!

Maximum viscosity at the lowest working temperature: 50 mm²/s!

Maximum density of the thermofluid: 1 kg/dm³.
Possible thermoregulation range within the range of the planned minimum and maximum working temperature.

Do not use thermofluids with any of the additives ether, ester, strong mineral acids, oxidizing acids or amines. Do not use demineralized water, mineral water, sea water or CaCl brines₂

Compatibility with the materials used for the thermostat (stainless steel 1.4301 (V2A) and with all the materials used in the system connected to the thermostat.

For a selection of thermofluids including technical data, please refer to the topical Huber catalog.

Safety Devices

Please Note:

The thermostat is equipped with devices that protect the thermostat and ensure operating safety and operational reliability:

Over-temperature protection device / overheating protection (3.3. , 4.4.):
This is an independent device that protects the thermal fluid from exceeding a user-entered temperature.

Low liquid level protection (4.1.):
A safety cut-out feature in the device that shuts down the device before the pump runs dry, e.g. in the case of hose rupture.

Alarm functions (4.3.8 - 9.):
Electronic monitoring of the temperature, pressure and electric current.

Hazards during thermoregulation

Caution! Potential risk of injury and material damage during thermoregulation:

In the course of operation of the thermostat, extreme changes in temperature and pressure and the specific characteristics of the thermal fluids used may constitute hazards.



Please Note:

All the safety instructions are vital and must be considered on the job in compliance with the present operating instructions.



3. Commissioning

3.1. Safety Instructions and Principles

Hazards through emission of fluids

Caution! Potential risk of injury and material damage:

The floor will be slippery when fluids have been spilt!
Thermal fluids with a low flash point constitute a fire hazard!
Hazard of scalding/burning when touching exposed or defective connections that are hot.

Prevent overflow of the sight glass. (A site glass is not fitted to open-bath units. This refers to the UCO range and other Chillers)
Prevent leaking fluid-conveying pipes/tubes and connections.
Always remove any liquids spilt on the floor immediately.
Always clean contaminated devices immediately.
Place an appropriately large collecting tray under your external application.
.

Hazards through emission of gases

Caution! Potential risk of injury and material damage:

Risk of causticization of your respiratory tracts and skin through vapors!
Prevent leaks on closed external devices.
Ensure good aeration and ventilation in the vicinity of the thermostat.
Choose thermal fluids for thermoregulation that are not detrimental to health.

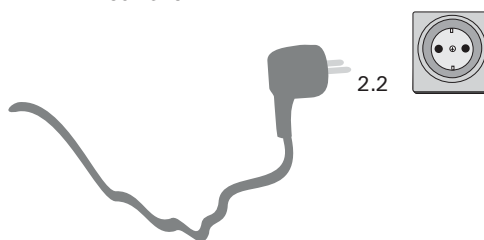
Current connection

- 2.1 Check the fuse, power and voltage ratings according to the data sheet (attached) and the type plate (on the rear side of the device)

- 2.2 Connect the power plug to the power outlet.

Shock-proof plug* for single-phase alternating current.

* Figure applies in the country of manufacture (Germany) only.



Please Note:

All the safety instructions are vital and must be considered on the job in compliance with the present operating instructions.



3. Commissioning

3.2. Preparing the thermostat for use

Preparation Thermo- regulation

If you wish to thermoregulate internally, please take the following into consideration:

Seal the pump manifolds with blind plugs and cap nuts (M16x1 / SW19). While doing so, counter using SW17 at the pump manifold.

If you wish to thermoregulate externally, please take the following into consideration:

Remove the blind plugs and cap nuts at the pump manifolds.
Replace them with suitable hose connections to your external application.

For more information, please refer to Chapter 4 „Thermoregulation via Controller“.

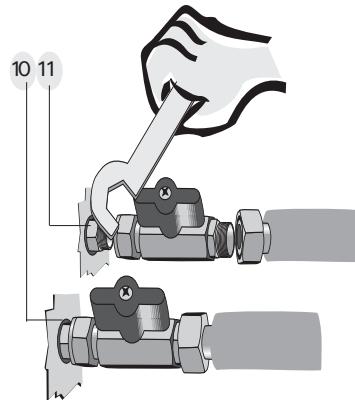


Alternative

Hoses can be connected via shut-off valves.
Only close when performing work on the reactor, otherwise always keep open!

Hazard!

Remember that thermal fluid expands and contracts with changes in temperature. Sealing the external application will expose the application to these forces!



Verify the following:

Make sure that all connections are correct and that there are no leaks!



3. Commissioning

3.3. Filling Thermofluid

Overtemperature protection

Requirements

Prepare the thermostat for thermoregulation and take safety measures as described below.

Setting the overtemperature protection device.

Requirements:

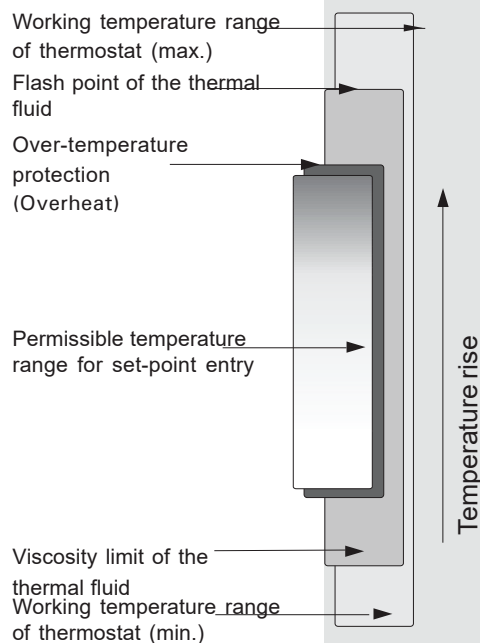
A suitable thermal fluid has been selected for the process requirements. The flash point of the thermal fluid is known.

Procedure:

The over-temperature protection is set to at least 25 °C below the flash point of the thermofluid.

Caution!

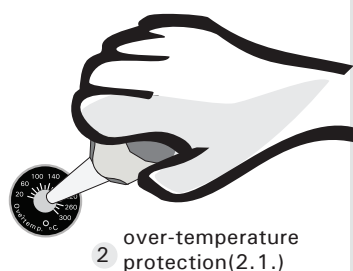
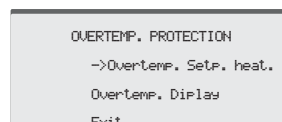
The overtemperature protection is an especially important safety device of your thermostat. It should always be operable and be tested periodically!



Setting the over-temperature protection

2

For the ministats 125, 230, 240 the overtemperature protection is set electronically. It is independent of the controller. Use a suitable tool (screw driver or the like) to press the button in the center of the over-temperature protection device. The overtemperature menu will be displayed.



Overtemperature Setpoint Heating: overtemperature value (adjustable on the Polystat controller 22-1).

Overtemperature Display: the actual overtemperature value is displayed.

For detailed information on the range of accessories incl. technical data and price quotations, please refer to the Huber catalog or contact your Huber agent.



3. Commissioning

3.3. Filling with Thermofluid

Bath thermostats

Filling the bath

- L2 Lift the bath cover
Fill in thermofluid.

Please note the optimum fill level:

with ministat 125, 10 mm below the bath opening.

with ministat 230, 25 mm below the bath opening.

with ministat 240, 25 mm below the bath opening.

Close the bath cover

Please note that the medium needs to cool to room temperature before you *replenish* thermofluid!

L2



Caution! Potential risk of injury and material damage!

In the case of high temperatures, the bath cover and housing cover become very hot.
Please touch the device and the cover at the grips only. **Scalding hazard!**

Never, under any circumstances, lift the cover of the bath during operation at high temperatures:

Scalding / burning hazard due to overflow of the thermofluid.
Risk of causticization of your respiratory tracts and skin through vapors!



Important!

For information on thermofluids, refer to 3.1! For a selection of thermofluids including technical data, please refer to the Huber catalog.



3. Commissioning

3.4. Major Presettings

Language

Deutsch

I When delivered the controller displays will be in German.
Other options can be selected in the „Language“ menu (refer to 4.3.2): English, Francais

Mains Failure Auto

Off

I After mains failure, manual input is required to continue the thermoregulation procedure.
Other options can be selected in the „Mains Failure Auto“ menu (refer to 4.3.7): On (enabled). After mains failure or even after activation/deactivation, the thermoregulation procedure will be continued automatically. The setpoint programmed last will be used for thermoregulation.

Compressor Mode

I Three options are available. (The default setting is „Auto“)
1. „Auto“ 2. „On“ 3. „Off“
1. „Auto“ means that the compressors will only operate when cooling is required.
2. Always Off means that the compressor will not turn on even if cooling is required
3. Always On means continuous compressor operation even when cooling is not required)

Control Mode

Internal control

I The thermal fluid temperature measured inside the thermostat is used as the controlled variable for thermoregulation.
Other options can be selected in the „Control Mode“ menu (refer to 4.3.4): Process control
Process thermoregulation (cascade) is carried out via a process Pt100 sensor placed at a measuring point outside of the thermostat, e.g. in reactor

Set-point Default set-point 20 °C

I The thermostat controls the temperature to the predefined setpoint of 20 °C.
Use the SET key and the encoder to select other options (refer to 4.1 and 4.7): Any temperature value within the performance parameters of the thermostat (refer to the Data Sheet 2.4 Appendix, type plate!) and the safety limits (refer to 3.1!).

Min. set-point Default minimum set-point 5 °C

Max. set-point Default maximum set-point 35 °C

I With this lower limit set at 5 °C it is not possible to enter a lower set-point.
The setting can be changed in the „Alarm Config.“ menu (refer to 4.4.1): Any temperature value within the performance parameters of the thermostat (refer to the Data Sheet, Appendix or type plate.) and the safety limits (refer to 3.1!).

I With this upper limit set at 35 °C it is not possible to enter a higher set-point.
The setting can be changed in the „Alarm Config.“ menu (refer to 4.4.1): Any temperature value within the performance parameters of the thermostat (refer to the Data Sheet, Appendix or type plate) and the safety limits (refer to 3.1).



All the parameters changed by the customer can be reset to the default values via the „Factory Default“ menu option.
The programs for operating the thermostat are described in Chapter 4 of the Operating Instructions.

4

Thermoregulation via controllers





4. Thermoregulation via controller

4.1. Safety Instructions and Principles

Checks to be Performed:

Test Arrangement



Safety Measures

Compare the device configuration, the system structure and the selected thermofluid to the thermoregulation target.
Verify the stability of the thermostat and external devices.
Make sure there are no leaking connections.
Make sure the shut-off valves for thermofluid and cooling water (if applicable) have been opened.
Check the connection to the power supply.

Caution! Potential risk of injury and material damage:

Slip hazard! The floor and the workspace may be slippery when fluids have been spilt!

Tilt hazard! Make sure the thermostat and external devices are in a stable position.

Shock hazard! Make sure the connection to the power supply is undamaged and in perfect working order.

Scalding and burning hazard! Always be aware of extreme temperatures.

Causticization hazard! Risk of causticization of your eyes, your skin and your respiratory tracts through hazardous vapors (depends on the thermofluid used).

Setting the overtemperature protection:

(Applies to Polystats and Compatible Control Thermostats):

When: Immediately after filling the system with thermofluid!

Recommendation:

Periodically test the function of the overtemperature protection by entering a higher setpoint, e.g. as follows:

Set the overtemperature protection to 30 °C.

Enter the maximum setpoint of 40 °C (since the overtemperature protection is independent of the controller, you can enter this excessive value without problems).

Enter the new setpoint of 33 °C (former setpoint 20 °C); the heating will heat to 33 °C, then the thermoregulation process will be stopped automatically with the error message: „Temp“

The error message will be displayed until the error has been remedied.

Remedy the error. Set the overtemperature protection to a temperature above 40 °C. Switch the thermostat off and then on again.

Entering the minimum and maximum setpoints:

In combination with the working temperature range limits, the minimum and maximum setpoints provide additional safety for the thermoregulation process. This means, accidental entry of a setpoint that is too low or too high will be rejected.

Low liquid level protection:

Monitor the liquid level during operation.

Applies to bath thermostats (Polystats and Compatible Control Thermostats):

Fill level to approx. 60 – 80 % of the bath height;

for chillers: fill level to approx. 60 – 80 % on the level indicator.

Thermofluid level too low: Risk of the thermostat pump running dry.

The controller will report an error and stop the thermoregulation process.

Thermofluid level too high: Overflow, soiling, slip and causticization hazard!

Change of fluid:

Rinsing fluid and thermofluids come into contact with stainless steel (V2A), Viton and Perbunan and must be compatible with these materials.

Room ventilation:

Sufficient aeration and venting in the vicinity of the thermostat minimizes the risk of overheating and the accumulation of harmful gases and vapors.



Please Note: All the safety instructions are vital and must thus be considered on the job in compliance with the present operating instructions.



4. Thermoregulation via Controller

4.1. Safety Instructions and Principles

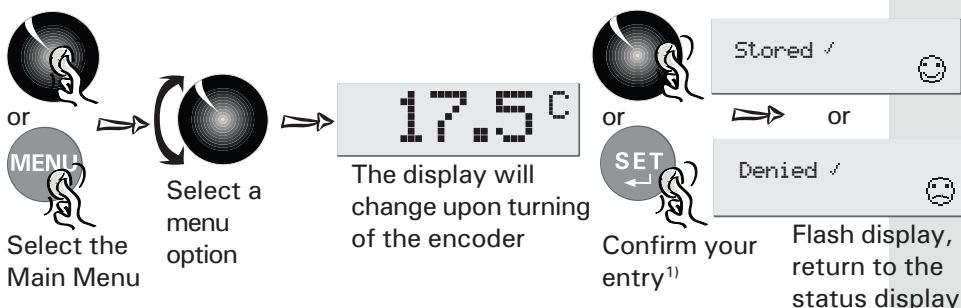
Principles of displays and entries

Salutation
Flash displays on the controller

Manufacturer	huber		
Salutation	Hallo		
System test	Electronics test / Memory Test ✓ Pump Test ✓	Poly cc 300 3 04.00a	Poly cc 300 -30.0 200.0
Status display	17.5 °C	Internal 17.5 °C Process n.a.	Internal 17.5 °C Process n.a. SetPoint 20.0 °C

The status display depends on the preset display mode (refer to 61-2).

Operating the controller



¹⁾ If you fail to confirm your entry within 4 minutes, your selection will not be saved. The program returns to the status display.
To exit the menu instantaneously at any point, use the **Break function***: Press the SET and MENU keys simultaneously. Your selection will not be saved.

Messages

During operation, ad-hoc messages may be displayed on the controller. They provide information on irregularities and hazards in the thermostat. In the case of imminent danger, the controller will display a message and stop the thermoregulation process/switch off the thermostat at the same time.

On/Off



****Press the On/Off key to start/stop the thermoregulation process (thermoregulation combined with circulation).**
For Ministat Control CC1 and CC2 during status display:
Turning the encoder displays the „Temperierung Ein/Aus“ (Thermoregulation On/Off) menu

Alternative:

****MasterClear function:** Pressing the MENU and TEMP keys simultaneously stops a thermoregulation program in progress or switches off the analog interface or switches from the digital interface remote mode to local mode.



Note: To learn more about the menu and the individual menu options, please also read sections 4.2 to 4.9 of the present operating instructions.

* applies to Polystat controller CC2 and CC3 only

** applies to Polystat controller CC3 only



4. Thermoregulation via Controller

4.2. Main Menu

Contents



The main menu provides menu options and submenu options including all the settings and selections required to operate the thermostat. (Refer to 42-1!)

¹⁾ This menu option is not available for Polystat controller cc1.

²⁾ This menu option is not available for Polystat controller cc2.

Main Menu

Submenus

Selection

MENU



Page	MAIN MENU:		Submenus
46-1	⇒ 2nd Setpoint ^{1), 2)}		
43-3	Alarm Clear		
43-3	Alarm Confis.		
43-5,6	Analogs-Interface ^{1), 2)}		
43-2	Display		⇒ Display
43-7	Disit. Interface ^{1), 2)}		⇒ Disit. Interface :
46-1	Ventine		
43-4	Machine Options		
43-4	Max. Heat Power		
45-1	Calibration Pros.		
43-4	Compressor Auto		
43-2	Mains Failure Auto		
45-2	Offset Calibration		
43-8	PI-Parameters		
48-2_5	Edit Program ¹⁾		
48-6	Program Start/Stop ¹⁾		
48-1	Start Ramp ¹⁾		
43-9	Acoustic Alarm		
43-4	Software version		
44-1	Setpoint Limits		
43-1	Language		
43-1	Temperature Scale		
43-1	Control Mode ^{1), 2)}		
46-1	Circulation		
43-9	Select Usermenu		
43-9	Confis Usermenu		
43-1	Factory Default		
43-2	Time Scale		
	Exit		
			⇒ ALARM CONFIGURATION:
			⇒ Alarm Mode
			Lower Alarm Limit
			Upper Alarm Limit
			Level Alarm Delay
			Exit
			⇒ Display
			⇒ Display modes
			optimise display
			Options
			Exit
			⇒ Disit. Interface :
			⇒ Hardware RS
			Baud rate
			Protocol
			slave address
			Exit
			I⇒ 2-P.CALIBR.:
			I⇒ Edit TCal1
			Edit TCal2
			Control to TCal1
			Control to TCal2
			Exit
			⇒ OFFSET CALIBRATION:
			⇒ internal sensor
			process sensor
			Exit
			⇒ INITIALISE:
			⇒ Unit Data
			User menus
			Programmer
			All together
			Exit



4. Thermoregulation via Controller

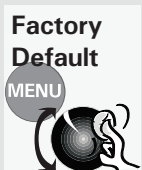
4.3. Editing Default Settings

1. Restoring the factory default
2. Selecting a language for the controller display
3. Temperature scale
4. Control mode – internal, process (cascade)



All the factory defaults can be customized in the menu.

All the factory defaults can be restored via the „Factory Default“ option (with the thermoregulation function switched off). For this purpose, proceed as described below. (Also refer to „Salutation“ 4.1).



```
MAIN MENU:
  Config User
  I->Factory default
```

„Factory Default“ option

```
INITIALISE:
  I->Unit Data
  Programmer
  User menus
  All together
  Exit
```

Submenu selection:

„Unit Data“: Important if you replaced device components or accessories.

„Programmer“

Deletion of all the thermoregulation programs

```
INITIALISE:
  User menus
  I->All together
  Exit
```

„User menus“

Restores the „User menu“ factory default.

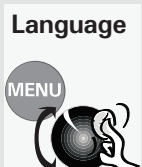
„All together“

Restores all the factory defaults.

Caution! All the thermoregulation programs incl. ramps entered by the customer will be deleted!

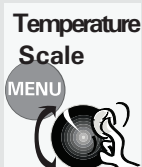
„Exit“ option:

The factory defaults will not be restored.



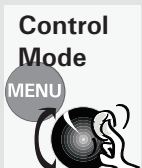
```
MAIN MENU:
  Set-point limits
  I->Language
  Temperature Scale
  Language:
  I->Deutsch
  English
  Francais
```

For selecting the language that is to be



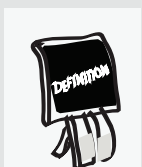
```
MAIN MENU:
  Language
  I->Temperature Scale
  Temperature Scale:
  I->Celsius
  Fahrenheit
  Kelvin
```

For selecting the temperature unit for display



```
MAIN MENU:
  Temperature Scale:
  I->Control mode
  Control mode:
  I->Internal
  Process (Cascade)
```

Not available for Polystat controller cc1!



For definitions of internal and external thermoregulation, please refer to Appendix 5.1. Huber Glossary, keyword Control Mode – Internal,



4. Thermoregulation via Controller

4.3. Editing Default Settings

5. Display
6. Time scale
7. Mains failure auto

All the factory defaults can be customized in the menu.

All the factory defaults can be restored via the „Factory Default“ option (with the thermoregulation function switched off). (Also refer to „Salutation“ 4.1).



Display



```
MAIN MENU:
  Disit. Interface
I> Display
  Edit Program
```

```
DISPLAY:
I> Display mode
  optimise display
  display
  Exit
```

```
DISPLAY MODES:
  Standard
I> Double
  Double1+Setp.
  Service1
  Service2
  exit
```

```
DISPLAY:
  Display modes
-> optimise display
  Exit
```

```
Please Enter:
Optimise display
```

„Display“ option

Selection in the „Display Modes“ submenu:
„Standard“:

Single-line status display, the actual value of the controller temperature (according to the control mode (internal or process temperature) is displayed in maximum font size.

„Double“:

Double-line status display, the actual values of the internal temperature and the (external) process temperature are displayed in medium font size.

„Double + Setp“:

Three-line status display, the actual values of the internal temperature and the process temperature as well as the setpoint are displayed in small font size.

„Service2“:

Four-line status display, the actual values of the internal temperature and the process temperature (external) as well as of the additional temperature sensors 1 and 2 are displayed in small font size.

Selection in the „Display Angle“ submenu:

Selection of a value to change the display angle by turning the encoder.

Time Scale



```
MAIN MENU:
  Temperature Scale
I> Time Scale
  Ventime
```

```
TIME SCALE:
  Seconds
I> Minutes
```

„Time scale“: option

Select the unit on which the timing of the thermoregulation programs is to be based.

Mains Failure Auto



```
MAIN MENU:
  PI-Parameters
I> PowerOff AutoStart
  Select Usermenu
```

```
POWEROFF AUTOSTART.:
I> Off
  On
```

„Mains Failure Auto“ option

„Off“ After mains failure, manual input is required to continue the thermoregulation process.

„On“ After mains failure, the thermoregulation process is continued automatically. The setpoint programmed last will be used for thermoregulation.

Refer to 4.1 Safety Instructions!



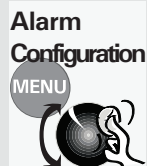
4. Thermoregulation via

4.3. Editing Default Settings

- 8. Alarm Configuration
- 9. Alarm Clear

All the factory defaults can be customized in the menu.

All the factory defaults can be restored via the „Factory Default“ option (with the thermoregulation function switched off). (Also refer to „Salutation“ 4.1).



```
MAIN MENU:
Alarm clear
I-> Alarm confis.
Analog Interface
```

```
ALARM CONFIG.:
I-> Alarm mode
Lower Alarm Limit
Upper Alarm Limit
Level alarm delay
Exit
```

```
ALARM MODE:
Run Mode
-> Stop Mode
```

```
Please enter:
Upper Alarm Limit
40.0°C
```

```
Please enter:
Level alarm delay
40
```

Selection of the parameters that are to trigger an alarm monitored by the software.

„Run Mode“

In the case of software-monitored alarm messages, the thermoregulation process is continued.

The alarm message will persist until the cause of the alarm has been eliminated and the alarm message has been acknowledged.

„Stop Mode“

The thermoregulation process will be stopped. Once the cause of the alarm has been eliminated and the alarm message has been acknowledged, it can be restarted manually.

Submenu selection:

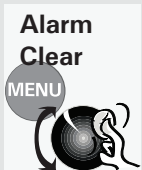
„Lower Alarm Limit“ / „Upper Alarm Limit“
The lower and upper alarm limits define the temperatures that trigger an alarm and stop the thermoregulation process, depending on the Alarm Mode settings.

Refer to 4.1 Safety Instructions!

„Level Alarm Delay“

The level alarm delay is defined by entering the delay time in seconds.

At commissioning, a delay of 60 seconds is recommendable to provide for uninterrupted distribution.



```
MAIN MENU:
Acoustic Alarm
I-> Alarm clear
Alarm confis.
```

„Alarm Clear“ option

Acknowledge the alarm.

```
ALARM-STOP MODE:
Please switch unit
off and on
```

„Alarm Mode“ option

„Restart“

The thermoregulation process is stopped and restarted. The message disappears.

```
ALARM CLEAR:
I-> Restart
Continue
```

„Continue“

The thermoregulation process is continued. The message disappears.



The following applies for each alarm mode:

Faults caused by hardware failures (overtemperature protection / overheating protection, low fluid level protection, pump winding thermostat) result in cut-out of the thermoregulation process.

Alarm messages due to hardware failures may be triggered by the overtemperature protection/overheating protection, by the low liquid level protection and by the pump winding thermostat.

Refer to 4.1 Safety Instructions!



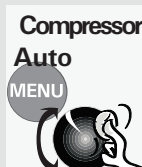
4. Thermoregulation via Controller

4.3. Editing Default Settings

- 10. Compressor Automatic
- 11. Maximum Heating Power
- 12. Software Version

All the factory defaults can be customized in the menu.

All the factory defaults can be restored via the „Factory Default“ option (with the thermoregulation function switched off). (Also refer to „Salutation“ 4.1).



```
MAIN MENU:
  Circulation
  I> Compressor Auto
  Config. Usermenu
```

```
COMPRESSOR AUTO:
  I> Automatic
  Always on
  Always off
```

Selection of the compressor starting mode

„Automatic“

The compressor will start automatically depending on the topical demand. The chiller will work on demand only.

Benefit: Saving of energy.

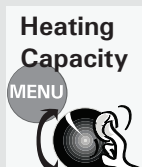
Drawback: Heating time (idle time) in the case of a sudden demand.

„Always On“

The compressor remains switched on, continuous chiller operation.

„Always Off“

The compressor remains switched of, the chiller is not in operation.



```
MAIN MENU:
  Machine Features
  I> Max. Heat Power
  Offset Calibration
```

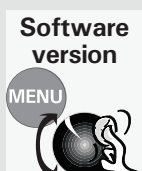
```
Please enter:
Max. heat power (%)
100
```

„Max. Heat Power“ option

Enter the desired maximum heating power in percent.

This is required for devices equipped with a heater and a compressor (chiller).

The heating power must be reduced to enable simultaneous operation of heater and compressor with a view to the fusing of the device.



```
MAIN MENU:
  Set-Point limits
  I> Software version
  Start/Stop Program
```

```
SOFTWARE:
Series No.: 0
Version   04.00s
Date     19.02.03. 15.41
```

„Software Version“ option

For example:

04.00s vom 19.02.2003, 15.41 PM.



4. Thermoregulation via Controllers

4.3. Editing Default Settings

13. Machine Features



Machine Features:

Various settings are possible depending on the equipment of the device. These can be selected in the submenu displayed. Only those options actually installed in the device are displayed.

MENÜ OPTIONEN:
I→ Pump Speed

Here, the nominal speed can be set for devices equipped with variable-speed pumps.

MENÜ OPTIONEN:
I→ Reservend

Without significance

MENÜ OPTIONEN:
I→ Exit

Settings are not changed, exit menu.

HAUPTMENÜ:
Language
I→ Machine Options
Max. Heat Power

„Machine Options“ menu option

Devices with signal interface according to the Namur standard

MENÜ OPTIONEN:
I→ Ext. Control Signal

Various external control signal actions can be triggered for devices with external control signal. For this reason, a submenu is displayed here for selecting the action when the external control signal is enabled.

MENÜ OPTIONEN:
I→ Ext. Control Signal
-Off-

„Off“ External control input is not assigned any function

MENÜ OPTIONEN:
I→ Ext. Control Signal
-Standby-

„Standby“ When the external control signal is enabled, thermoregulation is switched on and remains active until the external control input is disabled.

MENÜ OPTIONEN:
I→ Ext. Control Signal
-Act. 2nd Setpoint-

„Act. 2nd setpoint“: Switchover to the 2nd setpoint with actions of the external control input

MENÜ OPTIONEN:
I→ Ext. Control Signal
-Exit-

„Exit menu“: Settings are not changed, exit menu.



4. Thermoregulation via Controller

4.3. Editing Default Settings

14. Analog Interface – Parameter Input

Applies to all thermostats with the **Polystat Control cc3** !



All the factory defaults can be customized in the menu. All the factory defaults can be restored via the „Factory Default“ option (with the thermoregulation function switched off). (Also refer to „Salutation“ 4.1).



```

MAIN MENU:
Alarm mode
I> Analog Interface
Display

ANALOG INTERFACE:
I> Temp. T1 (Zero)
Temp. T2 (Span)
Conf. Input
Conf. Output
Exit

Please enter:
Temp. 1 (zero)
5.0°C

ANALOG INTERFACE:
Temp. T1 (zero)
I> Temp. T2 (Span)
Conf. Input
Conf. Output
Exit

Please enter:
Temp. 2 (Span)
35.0°C

ANALOG INTERFACE:
Temp. T1 (Zero)
Temp. T2 (Span)
I> Conf. Input
Conf. Output
Exit

ANALOG INPUT:
AIF-Input Off
AIF -> Set-point
I> Adjust
Exit

ANALOG INPUT:
I> AD-value at T1
AD-value at T2
On error Analoe
Exit

ANALOG INPUT:
AD-Value at T1
AD-Value at T2
I> On error Analoe
Exit

ANALOG INPUT:
AD-Value at T1
AD-Value at T2
I> On error Analoe
Exit

```

„Analog Interface“ (AIF) option: The thermoregulation process is controlled via an analog signal (currents from 4 to 20 mA), the strength of which represents the setpoint. The temperature range can be set by the user. The difference between *Zero* and *Span* must be at least 10 K but must not exceed 320 K.

Submenu selection: „Temp. T1 (Zero)“

Default „Temp.1“: Lower limit of the temperature range

Submenu selection: „Temp. T2 (Span)“

Default „Temp.2“: Upper limit of the temperature range

Submenu selection: „Param. Input“

Default: Parameter input disabled/enabled or configuration.

Submenu selection: „Configuration“

Default: Measured value of the analog-digital converter at T1/T2: „AN. CURR OK?- Yes“ if the analog device (provided by the customer) is synchronized with the Huber controller. „AN. CURR OK?- No“ if the analog device (provided by the customer) must be resynchronized with the Huber controller. Automatic return to the menu.

Selection in the „If Error at Analog“ submenu: Response to errors: Cut-out or enabling of the 2nd setpoint.



When the AIF is enabled, the input current determines the setpoint. If a setpoint is entered via the CC3 keyboard in this period, this setpoint will be enabled only after the AIF has been disabled. The definition of the setpoint via the AIF can be aborted with the MasterClear function. The setpoint defined prior to enabling the AIF will then be used for thermoregulation.

Caution! The controller module may be destroyed if currents exceeding 20 mA are used and/or if the polarity is confused!



4. Thermoregulation via

4.3. Editing Default Settings

15. Analog Interface – Parameter Output

Applies to all thermostats with the **Polystat cc3** controller!



All the factory defaults can be customized in the menu. All the factory defaults can be restored via the „Factory Default“ option (with the thermoregulation function switched off). (Also refer to „Salutation“ 4.1).



Analog Interface

MENU

```
MAIN MENU:
Alarm mode
I-> Analog Interface
Display
```

```
ANALOG INTERFACE:
Temp. T1 (Zero)
Temp. T2 (Span)
Conf. Input
I-> Conf. Output
Exit
```

```
ANALOG OUTPUT:
I-> Output value
Current at T1
Current at T2
Exit
```

```
SOURCE ANALOG OUT:
No output
Internal temp.
External temp.
Programmer
I-> Manual value
Exit
```

Select „Analog Interface“ (AIF): The thermoregulation process is controlled via an analog signal (currents from 4 to 20 mA), the strength of which represents the setpoint. The temperature range can be set by the user. The difference between *Zero* and *Span* must be at least 10 K but must not exceed 320 K.

Submenu selection: „Output Value“
Definition of the measuring point for temperature output values.

„No Output“ Constant output of 4 mA as a power supply, e.g. for an external thermometer.

„Internal Temp.“: Measurement using an internal sensor in the thermostat. In the case of a bath thermometer, the sensor is inside the bath.

„External Temp.“: Measurement with an external sensor, e.g. in an external bath.

„Programmer“: In a thermoregulation program, each segment can be assigned an output current, e.g. for adapting the speed of an external pump to individual program segments.

„Manual Value“ Any percentage from 0 %...100 % analog to 4...20 mA with encoder.

```
ANALOG OUTPUT:
Quelle fuer Ausgange
I-> Output value
Current at T1
Current at T2
```

```
Please enter:
DAC-Output at T1
591
```

```
ANALOG OUTPUT:
Output value
Current at T1
I-> Current at T2
Exit
```

```
Please enter:
DAC-Output at T2
4104
```

Submenu selection: „Current at T1“
Presetting of a converter output value for T1 (Zero): The encoder is used to set a value that results in the desired current value for T1, e.g. 4 mA at the analog setpoint encoder.

Submenu selection: „Current at T2“

Presetting of a converter output value for T2 (Span): The encoder is used to set a value that results in the desired current value for T2, e.g. 20 mA at the analog setpoint encoder.



Caution! The controller module may be destroyed if currents exceeding 20 mA are used and/or if the polarity is confused!



4. Thermoregulation via Controller

4.3. Editing Default Settings

16. Digital Interface

Applies to all thermostats with the **Polystat Control cc3!**



All the factory defaults can be customized in the menu. All the factory defaults can be restored via the „Factory Default“ option (with the thermoregulation function switched off). (Also refer to „Salutation“ 4.1).

Digital Interface MENU



```
MAIN MENU:
Control Mode
I-> Digit.Interface
Display
```

„Digit. Interface“ option

The controller is equipped with a bidirectional RS232 interface and an RS485 interface.

These digital interfaces enable remote control via a PC (Remote mode).

```
DIGIT. INTERFACE.:
I-> Hardware RS
Baud rate
Slave address
exit
```

Submenu selection: „Hardware RS“

Preselection of the RS232 (for 1 PC) or RS485 (for up to 32 PCs) interface.

```
HARDWARE RS:
I-> RS 232
RS 485
```

Submenu selection: „Baud rate“

```
DIG. SCHNITTST.:
Hardware RS
I-> Baud rate
Slave address
Exit
```

Preselection of the data transfer rate between the thermostat and the connected PC. You can select one of five baud rates.

```
BAUD RATE:
1200 Baud
2400 Baud
4800 Baud
I-> 9600 Baud
19200 Baud
```

Factory default: 9,600 Baud

Submenu selection: „Slave address“

```
DIGIT. INTERFACE:
Hardware RS
Baud rate
I-> Slave address
exit
```

The Huber thermostat is assigned an „address“, i.e. an assignment across the entire device system of the user. Selection range: 0 to 99.

```
Please enter:
Slave address
1
```



If the analog input is enabled as the setpoint source in the menu, then this setpoint has a higher priority than the setpoint sent to the controller via the digital interface.

Keyboard entries are not possible in remote mode. There is only one exception: the MasterClear function (press the MENU and TEMP keys simultaneously). In this case, the program exits the remote mode and the controller can be operated via the keyboard again. At the same time, the controller setpoint active prior to selecting „RS232“ or „RS485“ will be reactivated (auxiliary setpoint).



4. Thermoregulation via

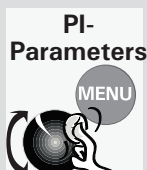
4.3. Editing Default Settings

17. PI-Parameters



All the factory defaults can be customized in the menu.

All the factory defaults can be restored via the „Factory Default“ option (with the thermoregulation function switched off). (Also refer to „Salutation“ 4.1).



PI-Parameters

Factory Default:

The P-parameter (proportional parameter) and the I-parameter (integral parameter) influence the thermoregulation behavior of your thermostat. The factory default of the P-parameter is 5,000, that of the I-parameter is 1,000.

The factory default is well suited for a large number of applications.

New settings:

The PI-parameters can be edited as desired.

Value range of the P-parameter: 50 ... 30,000

Value range of the I-parameter: 0 ... 30,000

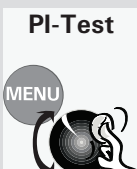
Test your thermoregulation process for optimum settings by entering new value pairs for the thermoregulation mode (internal or process)

Test Part 1	External thermoregulation Temperature change by 20 °C after each setting up to U* (ballistic effect)				
P - portion	50	1000	2000	3000 ...	10.000 ...up to Ü*e.g. 12.000
I - portion	0	0	0	0	0
Test Part 2	External thermoregulation Temperature change by 20 °C after each setting up to O**				
P - Portion	12.000	12.000	12.000	12.000	
I - Portion	1.000	2.000	5.000 ...	up to O**	e.g. 15.000

* U: Ballistic effect: At the thermoregulation target, the actual value oscillates about the setpoint.

**O: Optimum ratio between the accuracy of the controller and your desired speed.

Rule: Fast thermoregulation due to a high P and a low I-parameter results in a high ballistic effect.



PI-Test

MAIN MENU:
Offset Calibration
I> PI-Parameter
PowerOff AutoStart

Please enter:
P-Intern
5000

Please enter:
P-Internal
50

Please enter:
I-Internal
1000

Please enter:
I-Internal
0

Thermoregulation:

1. Heating by 20 °C with P/I = 50/0, then cooling by 20 °C
2. Heating by 20 °C with P/I = 1000/0, then cooling by 20 °C
3. Continue as described for Test Part 1 (table above) until all the relevant P-parameters have been tested.
4. Heating by 20 °C with P/I = 12,000/1,000, then cooling by 20 °C
5. Heating by 20 °C with P/I = 12,000/2,000, then cooling by 20 °C
6. Continue as described for Test Part 1 (table above) until all the relevant I-parameters have been tested.



4. Thermoregulation via

4.3. Editing Default Settings

15. Acoustic Alarm
16. Select Usermenu
17. Configure Usermenu

All the factory defaults can be customized in the menu.

All the factory defaults can be restored via the „Factory Default“ option (with the thermoregulation function switched off). (Also refer to „Salutation“ 4.1).



Acoustic Alarm



```
MAIN MENU:
  2nd St.point
I-> Acoustic alarm
  Alarm Clear
```

```
AACOUSTIC ALARM
I-> OFF
  ON
```

„Acoustic Alarm“ option
„Off“ Alarm signals and error messages without acoustic alarm.

„On“ Alarm signals and error messages with acoustic alarm.

Select Usermenu



```
MAIN MENU:
  PowerOff AutoStart
I-> select user menu
  Setpoint Limit
```

```
USER MENU:
I-> Administrator
  User menu 1
  User menu 2
  User menu 3
  User menu 4
  User menu 5
  User menu 6
  User menu 7
  Exit
```

„Select Usermenu“ option:

Only the administrator can select this submenu after entering the correct password.

Config Usermenu



```
MAIN MENU:
  Compressor Auto
I-> Config. User menu
  Control Mode
```

```
Please enter:
User menu
1
```

The „Config Usermenu“ option is visible to the administrator only (default see above „Select Usermenu“).

For this reason, only the administrator can select this submenu after entering the correct password.

Via the „Config. Usermenu“ option you define the menu options that are to be visible in the Main Menu, i.e. the factory default is customized.



The administrator password is forwarded to the user separately (on request).



4. Thermoregulation via Controller

4.4. Limiting the Thermoregulation Range

1. Setpoint Limits



All the factory defaults can be customized in the menu.

All the factory defaults can be restored via the „Factory Default“ option (with the thermoregulation function switched off). (Also refer to „Salutation“ 4.1).

Setpoint Limit



```
MAIN MENU:
  Select Usermenu
  I-> Set-point limit
  Software Version
```

„Setpoint Limits“ option:

Enter the desired minimum setpoint and confirm your entry (e.g. 10.0 °C).

```
Please enter:
Minimum Set-Point
5.0°C
```

The minimum setpoint is a safety limit for thermoregulation. Where:

The lowest permissible temperature value $\geq$ minimum setpoint (3.1, 4.1). It is not possible to enter a setpoint lower than the minimum setpoint.

```
Please enter:
Minimum Set-Point
10.0°C
```

Enter the desired maximum setpoint and confirm your entry (e.g. 170.0 °C).

```
Please enter:
Maximum Setpoint
35.0°C
```

The maximum setpoint is a safety limit for thermoregulation. Where:

The highest permissible temperature value $\leq$ maximum setpoint (3.1, 4.1). It is not possible to enter a setpoint higher than the maximum setpoint.

The maximum setpoint should not be set to a value lower than 5-8 degrees below the value set for the overtemperature protection. In this way, the controller can tolerate a minimum ballistic effect of the attained temperature when thermoregulating to the maximum setpoint.

```
Please enter:
Maximum Set-Point
170.0°C
```



Warning

The overtemperature protection is an especially important safety device of your thermostat. It should always be operable and thus be tested **periodically!**

If the actual value exceeds the set overtemperature limit, an alarm is triggered and the thermostat will cut-out the thermoregulation process. This process can be restarted only after the cause of the alarm has been eliminated and the alarm message acknowledged.

Refer to 3.1 Principles and Safety Instructions!



4. Thermoregulation via Controller

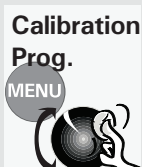
4.5. Calibration

1. Calibration Program



All the factory defaults can be customized in the menu.

All the factory defaults can be restored via the „Factory Default“ option (with the thermoregulation function switched off). (Also refer to „Salutation“ 4.1).



Calibration
Prog.

MENU

```
MAIN MENU:
  Analog Interface
  I> Calibration Prog.
  Circulation
```

```
2-P.CALIBR. INTERNAL:
  I> Edit T-Cal1/2
  Control to T-Cal1
  Control to T-Cal2
  exit
```

```
Please enter:
Set-point TCal1
      6 C
```

```
Please enter:
Set-point TCal1
     10 C
```

```
2-P.calibration:
  Edit TCal1
  I> Edit TCal2
  Control to T-Cal1
  Control to T-Cal2
  exit
```

```
Please enter:
set-point TCal2
     100 C
```

```
Please enter:
set-point TCal2
     40 C
```

```
2-P.calibration:
  Edit TCal1
  Edit TCal2
  I> Control to T-Cal1
  Control to T-Cal2
  exit
```

```
2-P.calibration:
  Edit TCal1
  Edit TCal2
  Control to T-Cal1
  I> Control to T-Cal2
  Exit
```

„Calibration Prog.“ option

Is used exclusively to calibrate the internal sensor.

Use a calibrated reference thermometer as a second temperature sensor for calibration.

Submenu selection: „Edit TCal1/2“

Entry of the 1st of two calibration temperatures, e.g. 10 °C.

Entry of the 2nd of two calibration temperatures, e.g. 40 °C.

Submenu selection: „Control to TCal1“

Start thermoregulating until the 1st calibration temperature has been reached.

If your reference thermometer indicates the set temperature reliably, compare the value to the actual value display of the Polystat controller. Deviations can be corrected using the encoder.

Submenu selection: „Control to TCal2“

Proceed as described for TCal1.

Exit the menu once you have completed the calibration procedure.



The two-point calibration includes all the values between T1 and T2. With offset calibration, on the other hand, the entire temperature level is shifted by one value on a linear level.



4. Thermoregulation via Controller

4.5. Calibration

2. Offset calibration



All the factory defaults can be customized in the menu.

All the factory defaults can be restored via the „Factory Default“ option (with the thermoregulation function switched off). (Also refer to „Salutation“ 4.1).



```
MAIN MENU:
Max. Heat Power
I-> Offset Calibration
PI-Parameters
```

```
OFFSET CALIBRATION:
I-> Internal sensor
Process sensor
Exit
```

„Offset Calibration“ option

Suitable for calibrating all the sensors used.

Use a calibrated reference thermometer as a second temperature sensor for calibration in the area of the sensor to be calibrated. Compare the actual controller temperature displayed to the reference thermometer reading.

```
Please enter:
Internal sensor
0.0°C
```

Any deviation can be corrected via the options provided in the „Internal Sensor“ submenu by entering the difference with the versed sign.

Example:

Controller display (Internal Sensor) 10 °C,
reference thermometer reading 8.5 °C,
difference = 1.5 K.

Calibrate by entering „-1.5“. The output level of the controller display will decrease by 1.5 K. The controller display will now correspond to the reference thermometer reading.

```
Please enter:
Internal sensor
-1.5°C
```

```
OFFSET CALIBRATION:
Internal sensor
I-> Process sensor
exit
```

Submenu selection: „Process Sensor“

Calibrate the process sensor (e.g. in an external bath)

Proceed as described for the „Internal Sensor“.



Offset calibration shifts the entire temperature level by one value on a linear level.

Two-point calibration, on the other hand, includes all the values between T1 and T2.



4. Thermoregulation via Controller

4.6. Utilities

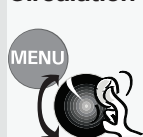
1. Circulation
2. Venting
3. 2. Setpoint (for Polystat Control cc3 only)



All the factory defaults can be customized in the menu.

All the factory defaults can be restored via the „Factory Default“ option (with the thermoregulation function switched off). (Also refer to „Salutation“ 4.1).

Circulation



```
MAIN MENU:
  Calibration Pros.
  I-> Circulation
  Compressor Auto
```

```
CIRCULATION:
  I-> Off
  On
```

„Circulation“ option

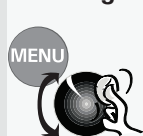
„Off“

No pump operation (related to the thermoregulation process) or stop of the alternating pump operation.

„On“

Start of the alternating pump operation (without thermoregulation) e.g. to enhance the filling procedure.

Venting



```
MAIN MENU:
  Time Scale
  I-> venting
  Exit
```

```
VENTING:
  I-> Off
  On
```

„Venting“ option:

Can be selected only with thermoregulation switched off.

Using the venting option, the pump can be operated in intervals in alternating mode, e.g. for enhanced venting of external applications.

„On“

The default settings for the time intervals for pump operation/pause may have to be edited (take into consideration the viscosity of the thermofluid and the system dimensions) and – at the same time – start of the alternating pump operation in intervals.

```
Please enter:
Pump ON (s)
10
```

```
Please enter:
Pump ON (s)
15
```

```
Please enter:
Pump OFF (s)
10
```

```
Please enter:
Pump OFF (s)
15
```

„Off“ Stop the alternating pump operation in intervals.

(Refer to 3.4!)

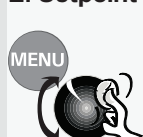


Caution! Potential risk of injury and material damage!

Please take into consideration the capacity and fill level of the thermostat and the connected systems as well as the viscosity and expansion characteristics of the thermofluid used.

Please prevent overflow of the fluid. Refer to 3.4!

2. Setpoint



```
MAIN MENU:
  I-> 2. Set-Point
  Acoustic Alarm
  Alarm Clear
```

```
Please enter:
2. Set-Point
15.0°C
```

```
Please enter:
2. Set-Pointt
25.0°C
```

„2nd Setpoint“ option (*exclusively available for Polystat controllers cc3*)

Entry of the 2nd setpoint. This setpoint is enabled only if an error occurs in the analog control. Refer to „Analog Interface“!

When entering the 2nd setpoint, the same applies as to the „standard“ setpoint: the characteristics of the thermofluid, the thermoregulation objective and the safety measures must always be taken into consideration.



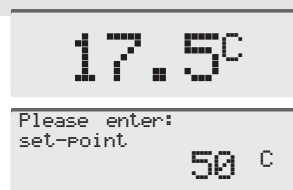
4. Thermoregulation via Controller

4.7. Enter a Setpoint - Start



All the factory defaults can be customized in the menu.

All the factory defaults can be restored via the „Factory Default“ option (with the thermoregulation function switched off). (Also refer to „Salutation“ 4.1).



Press the SET key.

The setpoint will be shown on the status display.

Enter the new setpoint by turning the encoder (e.g. to 50 °C).

Confirm your entry by pressing the encoder or the SET key.

Start thermoregulation only after all the prerequisites have been met (refer to 3 Commissioning), especially:

- Suitable location (3.1)
- Correct connections (3.2)
- Ambient temperature max. 30 °C (3.1)
- Correct setting of the overtemperature protection (4.1)
- Correct setting of the setpoint limits (4.4)

Polystat cc1 and cc2 controllers:

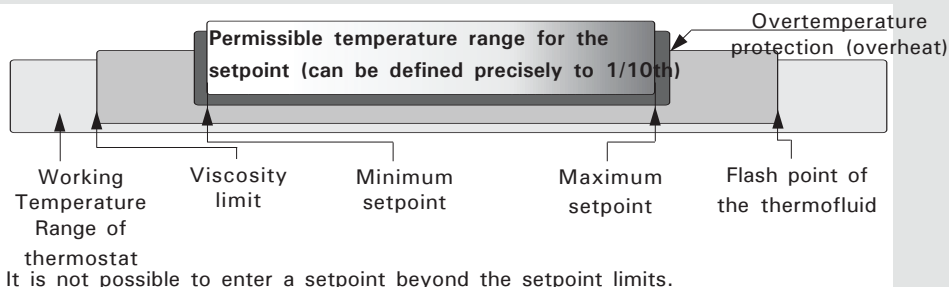
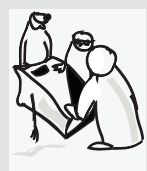
Start the thermoregulation process by turning the encoder.

Polystat cc3 controller:

The thermoregulation process to the new setpoint is started by pressing the TEMP key.



To stop the thermoregulation process: Refer to section 4.9!



Caution!

The overheating point and the setpoint must be 25 degrees below the flash point of the thermofluid and the setpoint must be above the temperature at which the thermofluid attains a viscosity higher than 50 mm²/s. (3.1.1)

Note: Gas venting (Prog. 55) and decalcify.





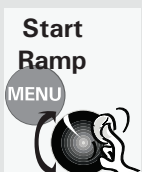
4. Thermoregulation via Controller

4.8. Convenient Thermoregulation – Programs

1. Start Ramp (for Polystat Control cc2/cc3 only)

All the factory defaults can be customized in the menu.

All the factory defaults can be restored via the „Factory Default“ option (with the thermoregulation function switched off). (Also refer to „Salutation“ 4.1).



If you want to change the working temperature slowly and smoothly instead of suddenly, you should implement the setpoint change via a ramp.

```
MAIN MENU:
Start/stop Program
I> Start Ramp
Temperature Scale
```

```
Please enter:
Go to temperature
20 C
```

```
Please enter:
Go to temperature
70 C
```

```
Please enter:
Time (min)
1
```

```
Please enter:
Time (min)
90
```

„Start Ramp“ option:

Enter the desired final temperature of the ramp (ramp setpoint), e.g. 70 °C.

Enter the time (in minutes) the thermoregulation to the ramp setpoint should take, e.g. 90 minutes.

Starting the ramp:
The ramp will automatically be started once you have confirmed the time parameter.

Completion of the ramp:
Once the ramp setpoint has been reached (e.g. after 90 minutes), the thermostat will keep the new actual temperature constantly on the new value (e.g. 70 °C).

Interrupting the ramp:
The ramp can be interrupted in the course of the process by entering a new setpoint (SET/encoder).



Caution!

The setpoint must be 25 degrees below the flash point of the thermofluid and above the temperature, at which the thermofluid attains a viscosity higher than 50 mm²/s. (3.1.1)



4. Thermoregulation via Controller

4.8. Convenient Thermoregulation – Programs

2. Edit Program (Exclusively for Polystat Control cc2/ cc3)

All the factory defaults can be customized in the menu. All the factory defaults can be restored via the „Factory Default“ option (with the thermoregulation function switched off). (Also refer to „Salutation“ 4.1).



Edit Program



Creation of a separate thermoregulation program

Polystat cc2 controller: 1 thermoregulation program for editing (can always be overwritten) with as many as 5 segments.

Polystat cc3 controller: 10 thermoregulation programs for selecting and editing (can always be overwritten). A total of 50 segments are available.

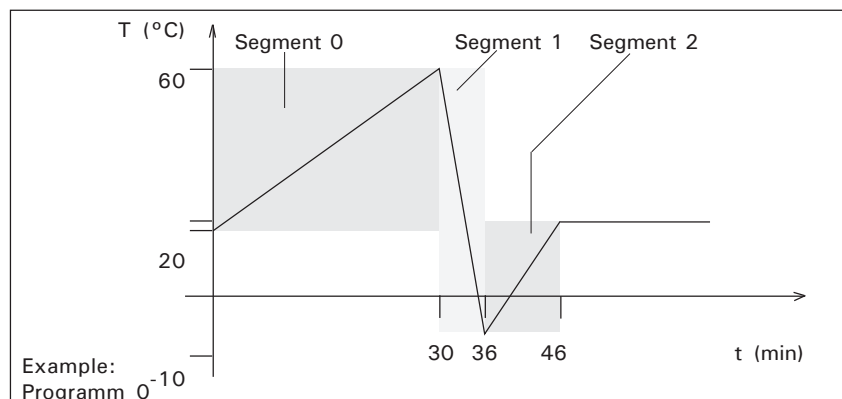
One segment can maximally span a period of 54 hours.

Programming is done in steps according to the „Edit Program“ menu item.

Planning (Example):

A fluid in an external bath is to be heated and cooled in 3 steps. The heating process is to be temperature-stable and the cooling process time-stable. For the time it takes to cool the bath, an agitator is to be controlled via the potential-free contact.

At the end of the program, the thermostat is to maintain the bath



```
MAIN MENU:
  PI-Parameters
  I-> Edit Program
  Start Progr.
```

```
PROGRAMMER:
  I-> Program 0
  Program 1
  Program 2
  Program 3
  Program 4
  Program 5
  Program 6
  Program 7
  Program 8
  Program 9
  exit
```

„Edit Program“ option

Options in the „Programmer“ (PR) submenu:
10 Programs

„Program 1“ option

Upon initial commissioning, all the programs will still be „empty“, i.e. they have not been assigned any segments. In the course of the application, these programs may be completely filled with segments.

For further programs, you may edit existing ones.



Caution!

The setpoint must be 25 degrees below the flash point of the thermofluid and above the temperature at which the thermofluid attains a viscosity higher than 50 mm²/s. (3.1.1!)



4. Thermoregulation via Controller

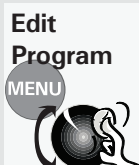
4.8. Convenient Thermoregulation – Programs

3. Edit Program (exclusively for Polystat Control cc2/ cc3)



All the factory defaults can be customized in the menu.
All the factory defaults can be restored via the „Factory Default“ option (with the thermoregulation function switched off). (Also refer to „Salutation“ 4.1).

Select „Exit“ to return to the superordinate level.



```
FUNCTIONS PR:
  Attach Segment
  I> insert Segment
  modify Segment
  Delete Segment
  show Segment
  Delete Program
  Exit
```

Submenu selection:

„Programmer“ (PR)/ „Program 2“/

„Functions PR“:

„Program 1“ options

„Attach Segment“ (for the 1st segment, this corresponds to „Insert Segment“). The parameters for the 1st segment are defined in the following.

(Refer to the example of program 1 on page 48-2!)

```
PROGRAM NO.1:
  I> Set point SegEnd
  Segment period
  Modify Segment
  Control mode
  Options
  Save & exit
  Exit
```

```
Please enter:
Set point SegEnd
0.0 C
```

```
Please enter:
Set point SegEnd
60.0 C
```

„Setpoint SegEnd“ option: entry of the setpoint for the 1st segment of the 1st program, e.g. 60 °C.

```
PROGRAM NO.1:
  Set point SegEnd
  I> Segment period
  Modify Segment
  Control mode
  Options
  Save & exit
  Exit
```

```
Please enter:
Segment period (s)
1
```

```
Please enter:
Segment period (s)
1800
```

„Segment period“ option: entry of the time period for the 1st segment of the 1st program, e.g. 1800 seconds.

```
PROGRAM NO.1:
  Set point SegEnd
  Segment period
  I> Control mode
  Options
  Save & exit
  Exit
```

```
CONTROL MODE:
I> Internal
Process (cascade)
```

```
CONTROL MODE:
Internal
I> Process (cascade)
```

„Control Mode“ option: example: selection of external thermoregulation.

```
PROGRAM NO.1:
  Set point SegEnd
  Segment period
  Modify Segment
  Control mode
  I> Options
  Save & exit
  Exit
```

```
OPTIONS PR:
  Pot.free Contact
  Analos Output
  I> End condition
  Stability
  exit
```

```
AT SEGMENT-END:
stop regulation
I> continue loop
```

„Options“ menu option: Selection of the „End condition“ for the 1st segment of the 1st program, e.g. „Continue“, i.e. the temperature value of the segment end is maintained. („Stop Regulation“ = thermoregulation to the former setpoint at the end of the segment)

```
STABILITY PR:
  Pot.free Contact
  Analos Output
  End condition
  -> Stability
  exit
```

```
STABILITY:
I> Time-stable
Temperature-stable
```

„Stability“ option, e.g. „Time-Stable“ for the 1st Segment of the 1st program.

Time-Stable The segment period entered has priority for the segment end.

Temperature-Stable: The „Setpoint SegEnd“ has priority for the end of the segment.



4. Thermoregulation via Controller

4.8. Convenient Thermoregulation – Programs

3. Edit Program

(Exclusively for Polystat Control cc2/ cc3)

All the factory defaults can be customized in the menu.

All the factory defaults can be restored via the „Factory Default“ option (with the thermoregulation function switched off). (Also refer to „Salutation“ 4.1).

Select „Exit“ to return to the superordinate level



Edit
Program

MENU



OPTIONS PR:
Pot.free Contact
Analog Output
End condition
Stability
I>exit

Select „Exit“ to return to the superordinate level „Program 1“.

PROGRAM NO.1:
Set point SesEnd
Segment period
Control mode
Options
I> Save & exit
Exit

„Save & Exit“ option: Saving of all the data for the 1st segment in the 1st program. The 1st program has thus been created.
If you exit the „Program 1“ level without „Save & Exit“, all the data entered for this segment up to now that have not yet been saved will be discarded.

After „Save & Exit“ the program will return to the superordinate level „Functions PR“.

FUNKTIONS PR:
Delete Segment
I> show Segment
Delete Program
Exit

Pros:0 Ses:0
Temp:60 Contr:Cas
Time:1800 Stab:Time
Poco:0 An0:0

„Functions PR“ option, „Show segment“ option: this is where you can check your entries. Press the MENU key to exit the display.

The program will return to „Functions PR“, where you select „Attach Segment“ as shown in the example. The entries now correspond to those for segment 1 taking the values from our example. In this segment, you must additionally select the potential-free contact for controlling the agitator in segment 2.

FUNKTIONS PR:
I> Attach Segment
insert Segment
modify Segment

Please enter:
Set point SesEnd
60.0

Please enter:
Set point SesEnd
-10.0 C

PROGRAM NO.1:
Set point SesEnd
I> Segment period
Control mode

Please enter:
Segment period (s)
1800

Please enter:
Segment period (s)
360

PROGRAM NO.1:
Set point SesEnd
Segment period
I> Control mode

CONTROL MODE:
I> Internal
Process (Cascade)

CONTROL MODE:
Internal
I> Process (Cascade)

Potential-free
contact

PROGRAM NO.1:
Control mode
I> Options
Save & exit

OPTIONS PR:
I> Pot.free contact
Analog output
End condition

POCO ACTIVE:
No
I> Yes

OPTIONS PR:
Analog output
I> End condition
Stability

AT SEGMENT-END:
I> Stop regulation
Continue loop
Repeat

OPTIONS PR:
End condition
I> Stability
exit

STABILITY:
time-stable
I> Temperature-stable

OPTIONS PR:
End condition
Stability
I> exit

PROGRAMM NO.1:
Options
I> save & exit
exit



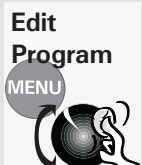
4. Thermoregulation via Controller

4.8. Convenient Thermoregulation – Programs

3. Edit Program (Exclusively for Ministat Control cc2/ cc3)



All the factory defaults can be customized in the menu.
All the factory defaults can be restored via the „Factory Default“ option (with the thermoregulation function switched off). (Also refer to „Salutation“ 4.1).
Select „Exit“ to return to the superordinate level.



Edit
Program
MENU

„Functions PR“ option, „Show segment“ option: this is where you can check your entries. Press the MENU key to exit the display.

The program will return to „Functions PR“, where you select „Attach Segment“ as shown in the example. The entries now correspond to those for segment 1 taking the values from our example. In this segment, you must additionally select the potential-free contact for controlling the agitator in segment 2.

FUNKTIONS PR: I-> Attach Segment Insert Segment Modify Segment	Please enter: Set point SesEnd -10.0	Please enter: Set point SesEnd 20.0 C
PROGRAMM NO.1: Set point SesEnd I-> Segment Temperiermodus	Please enter: Period Seem. (s) 360	Please enter: Period Seem. (s) 600
PROGRAMM NO.1: Set point SesEnd Period Seem. I-> Control mode	CONTROL MODE: I-> Internal Process (Cascade)	CONTROL MODE: Internal I->Process (Cascade)
	OPTIONS PR: Analog output I-> End condition Stability	AT SEGMENT-END: Stop regulation I-> Continue loop
	OPTIONS PR: End condition -> Stability exit	STABILITY: Time stable I-> Temperature stable
	OPTIONS PR: End condition Stability -> Exit	PROGRAMM NO.1: Options I-> save & exit Exit

You have now created a thermoregulation program with three segments according to our example.

Caution!

Before activating the program, make sure you are using a suitable thermofluid!

The setpoint must be 25 °C below the flash point of the thermofluid and above the temperature at which the thermofluid attains a viscosity higher than 50 mm²/s. (3.1.1)

Please Note:

You cannot create thermoregulation programs with the Polystat controller cc1.
You can create one thermoregulation program with the Polystat controller cc2.
You can create 10 thermoregulation programs with the Polystat controller cc3.





4. Thermoregulation via Controller

4.8. Convenient Thermoregulation – Programs

4. Start Program

(Exclusively for **Polvstat Control cc2/ cc3**)

All the factory defaults can be customized in the menu.

All the factory defaults can be restored via the „Factory Default“ option (with the thermoregulation function switched off). (Also refer to „Salutation“ 4.1).

For alarm messages, please refer to 4.3!



Start/ Stop Program

MENU



```
MAIN MENU:
  Software Version
  I> Start/Stop Program
  Start Ramp
```

```
PROGRAMMER:
  I> Program 0
  Program 1
  Program 2
  Program 3
  Program 4
  Program 5
  Program 6
  Program 7
  Program 8
  Program 9
  Exit
```

```
SERVICE-PROGRAM:
  Program Pause
  Program Continue
  Go To Segment No.
  Program stop
  Exit
```

```
Progr.:1      Seem.:0
Internal      21.2
Process       17.5
Setpoint      17.6
```

„Internal“:
development of the
internal actual value

„Process“:
development of the
external actual value

„Setpoint“: Calculated
topical setpoint,
development until the
segment end value has
been attained

60.0 °C

„Start/Stop Program“ option

If **no** program has been started:

Submenu selection:

„Program 0“ example Polystat cc3:

10 programs / 50 segments

(However one program cannot contain more
than 40 segments)

Polystat cc2: 1 program / 5 segments

If a program has **already** been started:

Submenu selection:

„Program Pause“

„Program Continue“

„Go To Segment No.“

„Program Stop“

„Exit“

Program 0 will be started.

Aborting the thermoregulation program

With Polystat controller cc2: Power switch or
new menu (using „Program Stop“).

With Polystat controller cc3:

Break function (press the MENU and On/Off
keys simultaneously).

Alternative: Power switch or new menu.

Standard end of the thermoregulation process
once the thermoregulation program has been
completely executed, according to:

- programmed segment end (the temperature
of the last segment setpoint is maintained
(Continue) or thermoregulation to the last
setpoint entered outside the program) or
repeating of Temp. Program.

- Stability: Time-Stable (i.e. after the
programmed segment period has elapsed) or
Temperature-Stable (i.e. after the segment
setpoint has been attained).

Once the program has been completely
executed, the status display will be shown.

Caution!

Before activating the program, make sure you are using a suitable thermo-
fluid!

The setpoint must be 25 °C below the flash point of the thermofluid and
above the temperature at which the thermofluid attains a viscosity higher
than 50 mm²/s. (3.1.1!)





4. Thermoregulation via Controller

4.9. Terminating the Thermoregulation Process

Entry New setpoint



The thermoregulation process is automatically continued once the setpoint has been reached, i.e. the thermostat keeps the temperature at the measuring point at the temperature defined by the setpoint. The thermoregulation process is continued until it is terminated manually.

This thermoregulation process can be changed at any time by entering a new setpoint (refer to 4.7).

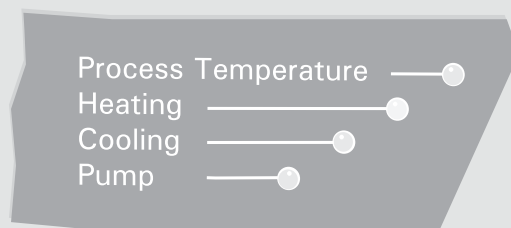
Abort



MinistatControl cc1/cc2: The thermoregulation process can be aborted at any time by pressing the power switch. You can also turn the encoder until the „Thermoregulation“ menu is displayed, in which you can select „Off“ or „On“.

Ministat Control cc3: The thermoregulation process can be aborted at any time by pressing the On/Off key on the controller.

Flow chart Refer to 2.3!



When you press the On/Off key or the power switch to stop the thermoregulation process, all the LEDs will go out.



Actuate the power switch of the thermostat to interrupt the power supply.



Caution!

Do not stop the thermoregulation process by pulling the power plug. When the controller is switched back on, various device messages may be displayed and faults may occur.

5

Shut Down





5. Shut Down

5.1. Safety Instructions and Principles



Caution! Potential risk of injury and material damage:

Slip hazard! The floor and the workspace may be slippery when fluids have been spilt!

Tilt hazard! Make sure the thermostat and external devices are in a stable position.

Shock hazard! Make sure the connection to the power supply is undamaged and in perfect working order.

Scalding and burning hazard! Always be aware of extreme temperatures. **Causticization hazard!** Risk of causticization of your eyes, your skin and your respiratory tracts through hazardous vapors (depends on the thermofluid used).

Electrical connections

Make sure that the power connections are in good condition. Plugs, sockets and cables must be in good order and of the correct rating. Avoid contact with fluids.

When replacing fuses, always use fuses with the correct ratings (refer to 5.3.!).

Cleaning

Clean and service the thermostat periodically (refer to 5.3!).
Keep the vicinity of the thermostat clean!
Always clean contaminated areas immediately.

Transport

Keep thermostats upright during transport.
Transport thermostats on clean, level and non-slip surfaces only.
Keep all the thermostat connectors/fittings closed.
Close the valves for fluids!
Protect plug-type connections using the caps provided for this purpose.



Please Note:

All the safety instructions are vital and must be considered during use in compliance with the present operating instructions!

5. Shut Down

5.2. Draining, Deactivating and Dismantling

Require- ments

Keep at hand a collecting vessel, a suitable drain tube, a screwdriver, compressed air and a cleaning cloth.

Terminate the thermoregulation process via menu selection or by pressing the On/Off key (R4 with cc3). (The final temperature should be as close as possible to the ambient temperature. Highly viscous thermofluids may need to be above ambient temperature).

Draining Thermo- fluid

Requirements:

Caution: Risk of burns! Before draining it, allow the thermofluid to cool or warm to almost room temperature. Set the temperature to a value slightly above room temperature for highly viscous media. Read the safety data sheet of the thermofluid in advance by all means; wear gloves, if required.

Avoid contamination in the vicinity of the thermostat through spilt or splashed thermofluid and through escaping hazardous vapors. Use appropriate drain tubes and collecting vessels.

Draining:

Stop thermoregulation, switch off the thermostat at the power switch. If the workspace in front of the thermostat is readily accessible, the device can be pulled directly to the edge of the workbench, and the thermofluid can flow freely into a collecting tank placed below the table, once the drain plug has been removed.

Otherwise, it can be drained via a suitable hose.

Put one end of the drain hose into the collecting tank.

Cautiously loosen the drain plug at the drain manifold. At the same time, have the end of the drain hose at hand.

Screw the drain plug completely out of the thread but still push it against the drain manifold. Remove the drain plug and at the same time slide on the hose.

The more rapidly and smoothly this is done, the less thermofluid is spilt. Complete draining can be supported in the end by lifting the device and slightly tilting it.

Switching off

Disconnect the power plug.

Cleaning surfaces

Cover unused electrical connectors using the protective caps supplied.

Clean the stainless-steel surfaces using steel cleaning spray (Huber catalogue number 6283).

Carefully clean painted surfaces using a mild detergent.

Note: The drained thermofluid must be disposed of in compliance with the instructions of the fluid manufacturer. The drained cooling water can be disposed of through the standard sewage.



5. Shutdown

5.3. Maintenance and Service

Requirements

Keep at hand a collecting vessel, a suitable drain tube, a screwdriver, compressed air and a cleaning cloth.

The thermoregulation process is terminated via menu selection or by pressing the On/Off key (R4 with cc3). (The final temperature should be as close as possible to the ambient temperature, with highly viscous thermofluids slightly above the ambient temperature).

Replacing Fuses



In the case of a failure of the automatic cut-out, determine the cause of the failure and eliminate it.
Then replace the fuse.

Cleaning the Surfaces

Cover unused slide-on receptacles using the protective caps supplied for this purpose.

Clean the stainless-steel surfaces using commercial steel cleaning spray. Carefully clean painted surfaces using brine made from a mild detergent.

Replacing the Controller



Caution! Disconnect the power plug before replacing the controller!!

If a defect comes up, you can replace the Ministat Control yourself. (When replacing the controller, please heed the safety instructions in the controller instruction manual.)

On the rear of the thermostat, there are two attachment screws. After you have loosened the two screws in the threads, apply uniform pressure to the screws to easily slide the slide-in module out to the front.

Caution! The bath cover must be closed!!

Urgent plea:

Maintain the warranty claim and the cost-efficient replacement price by not performing any repairs by yourself.



Caution!

The easiest and most effective method of preventing accidents and guaranteeing trouble-free operation:

Keep the vicinity of the thermostat clean!

Always clean contaminated areas immediately.

Service the equipment periodically!



5. Shutdown

5.3. Maintenance and Service

Maintenance of the Ministats

-  The liquefier is on the bottom of the ministat housing.
The cooling air is sucked in from below through the liquefier.
Avoid light particles in the air suction area on the bottom of the device.
-  The liquefier should be cleaned from time to time.
For this purpose, remove the power cable from the ministat,
completely drain the bath and tilt the housing to the rear by 90°. You can clean the liquefier using a brush or vacuum cleaner. Please
-  do not use any sharp or pointed objects for this purpose.
The ministat should be returned to its horizontal original position immediately after the cleaning procedure.

Disposal

-  To prevent damage to the environment, dispose of „veteran“ thermostats through certified expert companies only.

Service Hotline



Hotline Germany:	+ 49 - 781 - 9603 - 244
Hotline Service Center USA:	+ 1 - 740 - 373 - 6809
Hotline Service Center NL:	+ 31 - 485 - 542 - 811



Caution!

The easiest and most effective method of preventing accidents and guaranteeing trouble-free operation!
Keep the vicinity of the thermostat clean!
Always clean contaminated areas immediately.
Service the equipment periodically!

6

Appendix





6. Appendix

Programs
for changing
the factory
default settings

6.1. Presettings

Standard Parameter Set - Factory Default

4.3.8.	Alarm Configuration – Upper Alarm Limit	155 °C Ministat125
4.3.8.	Alarm Configuration – Upper Alarm Limit	205 °C Ministat230
4.3.8.	Alarm Configuration – Upper Alarm Limit	205 °C Ministat240
4.3.8.	Alarm Configuration – Lower Alarm Limit	-30 °C Ministat125
4.3.8.	Alarm Configuration – Lower Alarm Limit	-38 °C Ministat230
4.3.8.	Alarm Configuration – Lower Alarm Limit	-45 °C Ministat240
4.3.8.	Alarm Configuration – Level Alarm Delay	4 s
4.3.8.	Alarm Mode	Stop Mode
4.3.8.	Analog Interface – Temp. T1	5 °C
4.3.12.	Analog Interface – Temp. T2	35 °C
4.3.12.	Analog Interface – Parameter Input	disabled
4.3.12.	Analog Interface – Parameter Output /	
4.3.12.	Source Analog Out / Internal Temp.	
4.3.12.	Analog Interface – Current at T1	DAC 593
4.3.12.	Analog Interface – Current at T2	DAC 4104
4.3.5.	Display – Display Modes	Standard
4.3.5.	Display – Display Angle	100
4.3.13.	Digital Interface – Hardware RS	RS 232
4.3.13.	Digital Interface – Baud Rate	9600
4.3.13.	Digital Interface – Slave Address	1
4.6.2.	Venting	Off / 10 s
4.3.11.	Max. Heat Power	100 %
4.5.1.	Calibration Prog. – TCal 1	0 °C
4.5.1.	Calibration Prog. – TCal 2	100 °C
4.3.10.	Compressor Auto	Automatic
4.3.7.	Mains Failure Auto	Off
4.5.2.	Offset Calibration – Internal Sensor	0 °C
4.3.9.	PI-Parameters – P-Internal	2500 Ministat125
4.3.9.	PI-Parameters – P-Internal	1125 Ministat230
4.3.9.	PI-Parameters – P-Internal	1125 Ministat240
4.3.10.	PI-Parameters – I-Internal	1000
4.8.2.	Edit Program	0
4.4.1.	Acoustic Alarm	On
4.4.1.	Setpoint	20 °C
4.4.1.	Setpoint Limits – Minimum Setpoint	5 °C
4.4.1.	Setpoint Limits – Maximum Setpoint	35 °C
4.3.2.	Language	Deutsch
4.3.3.	Temperature Scale	°C
4.3.4.	Control Mode	Internal
4.6.1.	Circulation	Off
4.3.10.	Usermenu	Administrator
4.3.2.	Time Scale	Minutes
4.6.1.	2nd Setpoint	5 °C



All the factory defaults can be restored via the „Factory Default“ option (refer to 42-1).



6. Appendix

6.2. Interface Specification, Data



Please adhere to the following general principles:

- Always switch off all associated devices before connecting any cables.
- Only low voltages may be applied to the interfaces of the Huber devices.
- Always use an process sensor with a shielded cable. Otherwise the sensor may be electro-statically charged. The charge on the sensor could damage the measuring circuit when the sensor is connected.
- Before connection of plug-type connectors, ensure that they are in good and proper condition.
- Never try to make an electrical connection using force.

Interface
specification

Float switch
(level)

Pt100
external

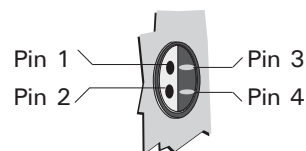
RS232/
RS485

Poko
dry contact
(Potential-
free contact)

ECS
External
control signal

Data
Communication

Pin 1: I +
Pin 2: U +
Pin 3: U -
Pin 4: I - (four-conductor technology)
Refer to 2.1, 2.2 and 3.5!



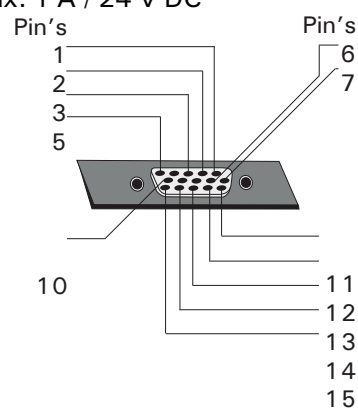
Pin 1: Terminating resistor 120 Ω, RS485
Pin 2: Data input RS232 (RXD)
Pin 3: Data output RS232 (TXD)
Pin 4: Reserved, please do not use!

Pin 5: Earth (DGND)
Pin 6: Potential-free contact (Poko) max. 1 A / 24 V DC

Pin 7: Terminating resistor
120 Ω, RS485

Pin 8: Reserved, do not use!
Pin 9: Reserved, do not use!
Pin 10: Potential-free contact (Poko)

Pin 11: A (RS485)
Pin 12: Analog output for AIF
Pin 13: Analog earth (AGND for AIF)
Pin 14: Analog input for AIF
Pin 15: B (RS485)



Refer to 2.1, 2.2 and 3.5!

Various protocols are available for complex parameterization to customize a Huber device to an automated application.

Examples: Point-to-Point commands for direct operation with a PC; LAI instruction set for RS485 bus systems, Modbus connection.

For more detailed information, please refer to: **Data Communication Manual**

ComBox According to the Namur Standard

6.3. Signal Interface

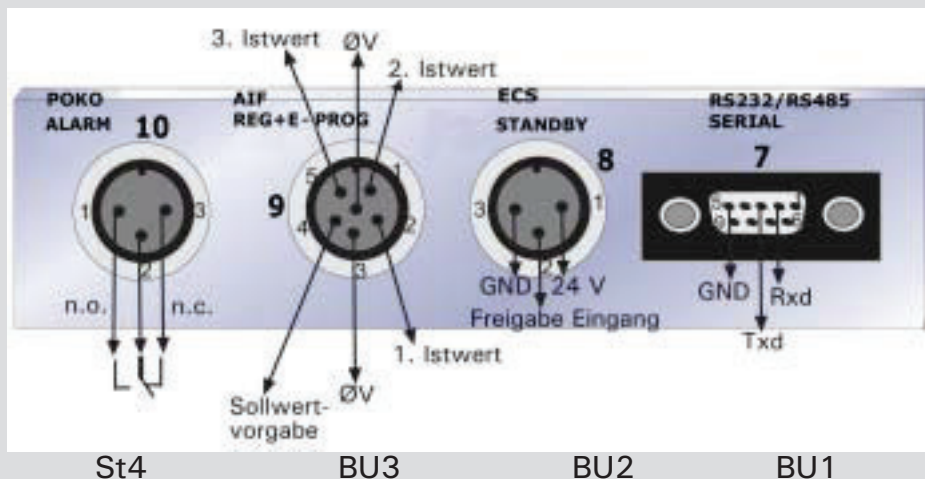
General Safety Principles

The CC-3 controllers are equipped with a 15-pin multi-function connector for connection to process equipment. For all Huber devices produced after August 2003 and equipped with a CAN connector, we offer an optional signal interface with an enhanced scope of functions. The signal interface communicates with the CC controllers via the internal CAN bus, independent of whether the device is operated with a CC-1, 2 or 3 controller.

Generally, the signal interface offers the following functions:

Only extra-low voltages are processed!

- Signal connectors are designed according to the NAMUR recommendations.
- Potential-free contact Break contact and make contact elements lead through.
- Enable signal can be implemented either via 24 V DC or with potential-free contact.
- Analog interface with one input (definition of the setpoint) and max. 3 output channels
- Digital RS232 interface with commands suggested according to NAMUR
- There is no possibility of confusing them

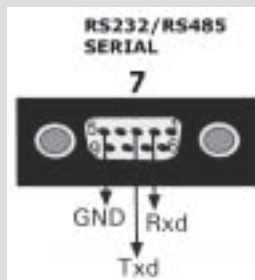


ComBox According to the Namur Standard

6.3. Electrical Connection Options

RS232/RS485, STANDBY/ECS

Warning: Never apply mains voltage to the plug connections of the signal interface! **Danger to life!**



RS232 Serial Interface (7)

A PC for remote control of the controller electronics can be connected to this socket.

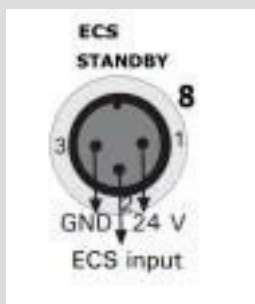
Before connecting the cable, check and adapt the settings (if required) in the „Digitale Schnittstelle“ (Digital Interface) menu.

RS232 circuit:

Pin 2	RxD	Receive Data
Pin 3	TxD	Transmit Data
Pin 5	GND	Signal GND

STANDBY input / ECS (8)

(for external emergency off and external control signal)



Pin	Signal
1	24 V / DC
2	ECS input
3	0 V

Use shielded cables only!

ACTIVATING THE STANDBY input:

* Select STANDBY

in the „Geräteeigenschaften – externes Steuersignal“ (Device Properties - External Control Signal) menu.

* Establish a connection to an external contact (AK, e.g. Emergency Stop) or an alarm contact of the superordinate system.

If the connection between Pin 1 and Pin 2 is interrupted by opening contact AK, or if an active enable signal (24V) is removed from Pin 2, the controller electronics will switch off the heater and the pump motor at all poles and switch to the „OFF“ status.

The „Pump“ LED will not light up for as long as the contact is open

.
When the contact is closed, the controller, consisting of the pump, the heater and the cooling system is enabled. The „Pump“ LED lights up.

ComBox According to the Namur Standard

6.3. Electrical Connection Options

STANDBY/ECS, analog signals

**Definition of the Setpoint through the Programmer:*

CC2 and CC3 only:

A program can be started with the contact closed; the „Pump“ LED and Heating/Cooling will light up. The program is processed until the contact is opened. Opening ends the program and the thermoregulation.

Closing of the contact later on causes a restart of the thermoregulation to the last setpoint that is influenced at program end depending on the configuration „Verhalten am Programmende“ (Behavior at Program End).

Caution: This is not a real Emergency Stop!

**STANDBY /
ECS** (external
control signal)

The controller queries the status of the enable signal, the action performed depends on the configuration..

Two types of controllers are provided:

1. A potential-free contact is provided, which is then applied to contacts 1 and 2.
2. An active, external enable signal with 0VDC or 24VDC is provided, which must then be connected to contact 2. Contact 3 is the connection to the 0 V root point of the external enable signal.

**Analog
Signals**

The signal interface features 4 analog channels. All channels have a common root point of 0V at contacts 3 and 6.

The analog input is connected to contact 4.

The analog input is used for feeding the setpoint for thermoregulation. This analog input and especially the temperature adjustment for fine calibration are handled via the „Analoginterface“ (Analog Interface) controller menu.

At contact 5, you can optionally output the setpoint, the internal actual value, the external actual value, a manually definable value or a programmable value can be output.

(Currently, the CC controller supports contact 5 only)

The signals are assigned to the analog outputs via the Controller menu.

The analog signals can be operated with 4...20 mA (0...20 mA when providing a corresponding load resistor, e.g. in the external wiring). The maximum load must not exceed 500 Ohm.

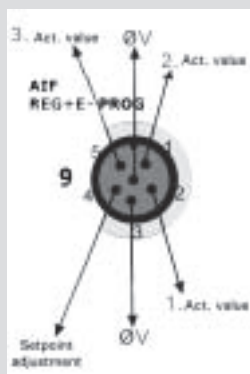


Safety Instructions for the Signal Interface Warning: Never apply mains voltage to the plug connections of the signal interface! Danger to life!

ComBox According to the Namur Standard

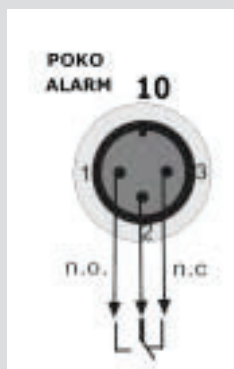
6.3. Signal Interface

AIF/REG + E-PROG, Alarm Output and Position Sensor



Programmer input/temperature recorder outputs (9) AIF/REG + E + PROG

Pin	Signal
1 Current output Channel1	4...20 V
3 Gnd for outputs	0 V
4 Programmer input EPROG	0...20 mA
5 Current output Channel3	0...20 mA / 4...20mA
6 Gnd for Programmiers	0 V



Alarm output and position sensor (10) (for external alarm signal)

This connection is designed in the form of a potential-free change-over contact. Pin 2 and Pin 3 are connected in the case of an alarm.

Breaking capacity	max. 30 W / 40 VA
Turn-on voltage	max. 125 V /-
Switching current	max. 1 A

Use shielded cables only!



Safety Instructions for the Signal Interface

Warning: Never apply mains voltage to the plug connections of the signal interface! Danger to life!

ComBox According to the Namur Standard

6.3. Digital Interface with Additional NAMUR Commands

RS232-, Command syntax, Namur commands

RS232 Commands

The ComBox features a digital RS232 interface.
According to the NAMUR recommendation (NE28), some commands are set up. The following character notation conditions apply if you use these Namur commands:

1 Start bit
8 character bits
1 Stop bit.
No parity

The transfer rate is set in the „Digitales Interface“ (Digital Interface) menu of the controller.

Data flow controller: Commands are not buffered, a new command may be sent as soon as the response to the topical query has been received. If a response it not to be expected, a break of 500 ms must be made. This restriction applies to the ‚Start‘ and ‚Stop‘ commands.

More notes on the command syntax

_ commands and parameters must be separated with a blank
_ Parameters are submitted with a floating point number or an integer
_ The full stop is the decimal separator in a floating point number (Code 46).
_ Following the comma, 2 characters are permissible in OUT commands.
_ A following physical unit is ignored.
_ A positive sign can be omitted.

Namur Commands

IN_PV_00	Query Tintern temperature (jacket)
IN_PV_02	Query Textern temperature (reactor inside temperature)
IN_SP_00	Query current thermoregulation setpoint
IN_SP_05	Query current analog setpoint
OUT_SP_00	Define xxx.xx setpoint
START	Activate thermoregulation
STOP	Deactivate thermoregulation
STATUS	Query device status

STATUS

Monadic

-1 Alarm

Manual Stop 0 OK / standby / manual stop

Manual Stop 1 OK Temp. or bleed



Safety Instructions for the Signal Interface

Warning: Never apply mains voltage to the plug connections of the signal

ComBox According to the Namur

6.3. Digital Interface with Additional NAMUR Commands

Command Sequence

Example

Command Sequence

Example of a Possible Thermoregulation Task:

Note: the notation \r\n means that CR LF is used as the terminating character of the command. The characters exchanged via the interface are enclosed in „“.

Master	Slave	Comment
„START\r\n“		Start thermoregulation
„OUT_SP_00 21.20\r\n“		Set setpoint to 21.2 °C
„IN_PV_00\r\n“		Setpoint queried
temperature)		Request Tintern (internal
	„20.5\r\n“	Slave sends the
temperatures		
„IN_PV_02\r\n“		Request Textern (external
temperature)		
	„20.5\r\n“	Slave sends the
temperatures		
„STOP\r\n“		Switch off thermoregulation

Note:

If the controller doesn't answer, a break of 0.5 s must be maintained.



Safety Instructions for the Signal Interface

Warning: Never apply mains voltage to the plug connections of the signal interface! Danger to life!



6. Appendix

6.4. Device Messages

„Device messages“ are codes that inform the user or service technician of „occurrences“ and the current status of internal components and systems at the time of the „occurrence“.

The controller display's device messages automatically

Device messages can be a warning or a fault.

All alarm messages shut down the thermoregulation process.

Factory default settings determine that all device messages are accompanied by an acoustic alarm. (This can be disabled via the „Acoustic Alarm“ option.)

All alarm messages will persist until the cause of the alarm has been rectified.

All alarm messages must be acknowledged via the „Alarm Mode“ option.

Once the alarm has been acknowledged and the cause of the alarm has been rectified, the alarm message will disappear.

Once the cause of the alarm has been rectified, the thermostat will function as programmed.

The display shows the alarm message in the form of a „flash“ display.

In intervals of 2 seconds, the display alternates with the status display.

„Alarm !!!“ is displayed in the upper line of the alarm message.

The cause of the alarm is displayed in the bottom line of the alarm message.

To acknowledge the alarm message:

Select the „Alarm Mode“ option. Display in Stop Mode (refer to 4.3.9) Follow the displayed prompts.

Display in Stop Mode (refer to 4.3.9)

Select „Restart“ or „Continue“.

The following applies to all automatic messages that are identified by the „Practice“ icon:

Please check the external conditions of your thermostat and your controller inputs according to the operating instructions and your process arrangement.

If required, re-program the condition.

If the message persists after you have repeated this action, please contact our Service Hotline.

The following applies to all spontaneous messages that are identified by the „Hotline“ icon:

Please switch off the thermostat and call our Service Hotline

Hotline Germany: (+ 49)781 – 9603 – 244

Hotline Service Center USA: (+ 1)740 – 373 – 6809

Hotline Service Center NL: (+ 31)- 485 – 542 - 811



ALARM !!!

32.0°C

ALARM MODE
switch on and off

ALARM MODE
I> Restart
Continue





6. Appendix

6.4. Device Messages



Description of the message.
Behavior of the thermostat.
Your response.
Refer to 63-1!

Hardware
error

device not
initialized

Cause

During the self diagnostics carried out by the machine after a controller has been replaced a new controller is detected by the operating system. This error just means that the system does not recognize the new controller.

Effect

The circulation and thermoregulation process cannot be started.

Remedy

*Please select the „Unit Data“ option in the „Factory Default“ menu.
This eliminates the error.*

Hardware
error

ALARM !!!

HARDWARE ERROR !!



Cause

During the self diagnostics carried out by the machine after a controller has been replaced a new controller is detected by the operating system. This error just means that the system does not recognize the new controller.

Effect

The circulation and thermoregulation process cannot be started.
Please call our customer service!

Device ID
error

ALARM !!!

Device ID error !!



Cause

During the self diagnostics carried out by the machine after switch-on an error is detected in the hardware or software. This could happen after a controller replacement.

Effect

The circulation and thermoregulation process cannot be started.

Remedy

Please call our customer service!

Over-
temperature

ALARM !!!

OVERTEMPERATURE !!



Cause

The temperature of the thermal fluid exceeds the temperature limit of the over-temperature protection device.

Effect

The circulation and thermoregulation process is stopped.

Remedy

Please check your set-point or your thermoregulation program as well as the setting of the over-temperature protection device and compare it to the safety requirements of your process and the application limits of your thermofluid.

Correct your inputs or the setting of the over-temperature protection (refer to 3.1, 3.4, 4.1, 4.4.1 and 4.7).

Restart the thermoregulation process according to the alarm mode.

Over-
temperature

ALARM !!!

ALARM TEMPERATURE !!



Cause

In Alarm Run Mode only (refer to 4.3.8.): The thermal fluid's temperature exceeds the „upper alarm limit“ or has dropped below the „lower alarm limit“.

The circulation and thermoregulation process is stopped.

Remedy

Please check your set-point or your thermoregulation program as well as the setting of the over-temperature protection device and compare it to the safety requirements of your process and the application limits of your thermofluid.

Correct your inputs or the setting of the alarm limits (refer to 3.1, 3.4, 4.1, 4.3.8 and 4.7).

Restart the thermoregulation process according to the alarm mode.



6. Appendix

6.4. Device Messages



Description of the message.
Behavior of the thermostat.
Your response.

Refer to 63-1!

Level

ALARM !!!

LEVEL !!



Cause

The level has dropped below minimum allowed thermofluid level or the circulation pressure of the pump is too low.

Result

The circulation and thermoregulation process is stopped.

Remedy

Please check the thermofluid filling level and if necessary, replenish the thermofluid, (refer to 3.4).

Restart the thermoregulation process.

If the alarm message persists, please check for visible causes of extraordinarily high fluid losses through loose or defective connections, defective tubing, fluid contamination etc. and rectify. If the alarm persists despite your remedial action, please call our customer service!

ALARM !!!

PUMP MOTOR !!



Cause

The circulation pump is overloaded and the pump motor has overheated.

Result

The circulation and thermoregulation process is stopped.

Remedy

Make sure that the thermofluid in use is suitable for the temperature range over which the thermostat is used.

The viscosity of the thermofluid must not exceed 50 mm²/s.

If required, change the thermal fluid.

Once you are sure that the correct thermal fluid is in use or replaced the thermal fluid with one that is suitable, allow the pump motor to cool and re-start.

If the alarm message persists and you are using a suitable thermal fluid, please call our customer service!

ALARM !!!

WT error chain !!



Cause

A general fault has been detected with the motor windings or an incorrectly rotating 3-phase supply has been detected. (A trained electrician must swap any two phases in the mains plug to change the direction of rotation of the 3-phase supply)

Effect

The circulation and thermoregulation process is stopped.

Remedy

Make sure that the thermofluid in use is suitable for the temperature range over which the thermostat is used.

The viscosity of the thermofluid must not exceed 50 mm²/s.

If required, change the thermal fluid.

Check the condition of all circuit breakers and re-set as necessary.

Once you are sure that the correct thermal fluid is in use or replaced the thermal fluid with one that is suitable, allow the pump motor to cool and re-start.

If the alarm message persists and you are using a suitable thermal fluid, please call our customer service.



6. Appendix

6.4. Device Messages



Hardware
error

ALARM !!!

PCP INIT ERROR !!

Description of the message.
Behavior of the thermostat.

Cause

During the self diagnostics carried out by the machine a hardware or software error has been detected. This can sometimes happen after fitting a new controller.

Effect

The circulation and thermoregulation process cannot be started.

Remedy

Please contact our customer service!

Hardware
error

ALARM !!!

INTERNAL SENSOR !!



Cause

An internal temperature sensor is not connected or is defective.

Effect

The circulation and thermoregulation process cannot be started.

Please call our customer service!

Device ID
Error

ALARM !!!

EXTERNAL SENSOR !!



Cause

External Pt100 sensor is not connected or defective.

Effect

The circulation and thermoregulation process cannot be started.

Remedy

Please check the connection of your process Pt100 sensor including the connecting cable and the sensor itself. If the sensor is defective, replace it with a new one.



6. Appendix



6.5. Huber Glossary

AIF – Analog Interface The analog interface is used to transfer constantly changing values, mostly temperatures, in the form of a signal current (generally 4 to 20 mA). Refer to „External Control“! Refer to 4.3!

Alarm Limit The defined absolute temperature for the thermofluid or product depending on the control mode. When the temperature exceeds or drops below the defined value, the thermostat will respond as defined in the alarm mode.

Alarm Mode

The response of the thermostat when the temperature has exceeded or dropped below the defined alarm limits.

Ambient Temperature Range

This is the temperature range permissible in the area surrounding the device for the device to function properly.

For all devices by HUBER, this is the temperature range from 5 to 30 °C. The cooling performance quoted is always based on an ambient temperature of 20 °C.

Automatic Mains Failure Control (Mains Failure Auto)

Determines the response of the thermostat in the case of a power failure.

The corresponding settings are made in the main menu of the controller.

Default setting:

Mains Failure Auto is disabled. Once the supply of power has been restored, manual intervention is required to continue the thermoregulation process.

Alternative setting:

Mains Failure Auto is enabled. Once the supply of power has been restored, thermoregulation is automatically continued. Refer to 4.3!

Automatic Cooling Capacity Control

A method for saving energy and resources. The microprocessor control determines whether the required cooling capacity can be reduced according to the operating temperature. The adaptation is performed continuously and – apart from saving energy – it also protects the compressor, decreases heat dissipation into the environment and reduces the consumption of cooling water.

Refer to Compressor Auto, 4.3!

Automatic Compressor Control (Compressor Auto)

Supports an energy-saving method for cooling thermostats in specific thermoregulation situations. A control logic in the microprocessor helps decide whether the chiller is needed at all or whether it can be switched off.

Refer to 4.2.! Refer to Automatic Cooling Capacity Control!

Actual Temperature

The temperature at the measuring point at the time of display. The measuring point is either in the thermostat, in the thermofluid (factory default: internal thermoregulation) or external to the thermostat, e.g. in the core of a reactor (external thermoregulation).

Bath, Closed

Double-wall, closed „shell“ with inflow and outflow connections.

Indirect thermoregulation of the customers' products in the „core“. Designed in metal or glass. Refer to „Reactor“, refer to 3.3!

Bath Opening

This is the active surface available for direct thermoregulation, generally across the entire usable depth.

Bath Thermostat

A bath thermostat is a thermostat equipped with a circulation pump (force pump in Polystats) and a bath tank for holding the object to be thermoregulated. The circulation pump is primarily used for mixing the bath fluid, if required, however, it can also pump the fluid through a closed external circuit, e.g. when flow water coolers for cooling heating thermostats are connected.

Bath Thermostat/Circulator

Thermostats with a bath opening large enough to accommodate objects for direct thermoregulation in the bath, including a circulation pump (force or suction pump with Compatible Control thermostats) for closed (force pump) or open external (suction pump) circuits.

Bath Volume (also referred to as fill volume)

The volume of the thermofluid inside the bath, required for the intended operation of the thermostat, excluding the thermofluid volume in external fluid circuits.

Especially in closed external applications, the expansion tank must be dimensioned with care since the circulator must additionally handle the expansion of the fluid in the external circuit.

Baud Rate

The data transfer rate for serial communication (mandatory setting).

Refer to: 4.2 Digital Interface!

Bath Tank

An open tank that holds the thermofluid (fluid for thermoregulation).



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6.5. Huber Glossary

Calibration Thermostat (CAL)

A bath thermostat with an especially high degree of temperature stability and especially uniform temperature distribution across the bath fluid.

Calibration

Verification of the measured values displayed and – if required – recording of the values by which the displayed values deviate from the

Cascade Control

Refer to Process Control!

Characteristic Temperature

This term denotes the operating temperature of a heating thermostat attained in stationary status with the heating switched off and the pump in operation. It depends on the motor capacity of the pump installed, the bath fluid used (viscosity, density) and the thermostat

Compressor

A compressor is a machine for compressing gases and vapors.

Condenser

Apparatus in chillers for condensing refrigerant vapors (liquefier).

Cooler (Special Feature of the Unistat Tango)

This type of cooler is a heat exchanger through which water flows during the cooling process. This cooling water leads off just enough heat from the superheated refrigerant to prevent condensation.

The cooling water may flow only while the compressor is in operation, otherwise the cooling unit will not work!

Cooling Capacity

Heat abstraction from the thermofluid via heat exchangers, using either ambient air or cooling water.

Cooling/Heating Thermostat

A thermostat with a working temperature ranging from below to above the ambient temperature

Cooling Thermostat

A thermostat, the working temperature range of which is generally below the ambient temperature and that preferably abstracts heat from the thermofluid. HUBER cooling thermostats are actually cooling/heating thermostats since their working temperature ranges from below to above the ambient temperature and they are

Cooling Water

Water routed through the thermostat via a system of pipes for cooling the chiller. It should contain only minimized amounts of lime and

Cooling Circulators (IC, UC)

These are cooling thermostats designed in the form of circulators. Due to their special design (desktop, tower, without a readily accessible bath), their cooling performance and pumping

Circulation Pump

The circulation pump is used to circulate the thermofluid in a closed circuit.

Circulator (Unistat)

A circulator is a thermostat, in which the thermofluid is routed through an open or closed external circuit. Unistats feature a thermically decoupled, active surface (expansion tank). In this case, however, the surface temperature

Circulation (operating mode)

Circulation means that the thermofluid circuit is driven by the circulation pump. The heater and/or cooling unit are switched off.

Controller

In a long tradition at Huber's, „controller“ refers to an exchangeable electronics unit, including

Control Mode – Internal, Process (Cascade)

Internal thermoregulation: The temperature measured inside the thermostat (flow temperature) is used as the controlled variable for thermoregulation. **Process (Cascade):** With the aid of an external Pt100 sensor, a measuring point outside the thermostat in an external application, e.g. in a reactor, is used for controlling the thermoregulation process.



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6.5. Huber Glossary

Default Parameters

Refer to Factory Default!

Delivery Pressure

The overpressure of the circulation pump of a thermostat directly at the pressure connecting piece, measured using water. On charts, the delivery pressure is expressed as a function of the flow rate.

Delivery Suction

The suction of the circulation pump (force/suction pump or duplex pump) directly at the pressure connecting piece, measured using water. On charts, the delivery suction is expressed as a function of the flow rate.

Digital Interface

The digital interface is used to transfer data in the form of bits and bytes.

DW-Therm

DW-Therm is a thermofluid (fluid for thermoregulation) developed especially for Unistats (Huber thermostats with a closed thermofluid circuit) with an extremely large temperature range (-90°C... + 200°C).

Effective Heating Capacity

Heat flow delivered to the thermofluid (fluid for thermoregulation) by heat sources.

Extended Working Temperature Range

This is the working temperature range extended to lower temperature levels achieved through the use of a manufacturer-defined cooling coil with cooling water.

External Control

The thermostat is not controlled through the controller but rather through external equipment. The external interface (multifunction socket) on the Polystat controller is used for this purpose. Refer to: 2.3, 4.3!

Factory Default

The manufacturer has set the thermostat parameters such that the temperature range constitutes only a minimum potential hazard and that the easiest and most probable thermoregulation process can be performed. Factory defaults ensure safe commissioning by the customer and can be changed using the respective programs according to the requirements of the customer. All the factory defaults can be restored with Program 52 with the thermoregulation function switched off. Refer to 4.2 Restart!

Fill Volume

Refer to Bath Volume!

FL

Refer to Safety Classification!

Flash Point

The term flash point denotes the temperature to which a fluid in an open bath must be heated until the vapor/air mixture at the surface inflames when a flame is held near and keeps on burning once the flame is pulled away again.

Refer to: 3.1.EN 61010-1 EN 61010-1

Flash Display

Controller display shown for seconds only to provide information on a variable intermediate program status or the status of the thermostat.

Flow Cooler (DC)

Flow coolers are „uncontrolled“ coolers without circulation pump that are interposed in an external circuit, thus expanding the functionality of a heating thermostat to that of a heating/cooling thermostat. On the one hand, they substitute water cooling, on the other hand they are used to achieve lower temperatures.

Flow Rate

The volume of fluid delivered by the circulation pump per time unit, measured using water. In charts, the flow rate is expressed as a function of the delivery pressure (backpressure).

Flow Temperature

Denotes the temperature measured directly at the point the thermofluid exits the circulator.

Force Pump

The force pump is used for circulating the thermofluid in a closed external circuit and for mixing the fluid inside the bath.

Force/Suction Pump

This pump features a pressure and a suction stage driven by a common motor. The pressure stage forces the thermofluid from the thermostat into the circuit, the suction stage draws the fluid back into the thermostat. A force/suction pump can be used in a closed circuit. Benefit: In the external circuit, the pressure drops from positive values (pressure) to negative values (vacuum) and is almost zero in the consumer. This facilitates thermoregulation even of pressure-sensitive glass vessels. In addition, an open external circuit (bath tank) can also be connected using a force/suction pump. A level constant ensures a constant fluid level in the bath tank.



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6.5. Huber Glossary

Definitions and Cross-References

Gas Venting

A special thermoregulation regime common to all Unistats with the objective of perpetuating thermoregulation despite the evaporation of part of the thermofluid and removing elements with a low boiling point from the circulating fluid.

Heat Exchanger

Refer to Thermofluid!

Heating Capacity

The maximum installed electrical power of the heating element. The heating capacity depends on the voltage of the thermostat. It is continuously controlled and reduced when the temperature approaches the programmed setpoint.

Heating Thermostat

A thermostat, the working temperature range of which is primarily above the ambient temperature and that preferably transfers heat to the thermofluid.

High-Pressure Cut-Out

Protective switch installed in Unistats and Unichillers. Shuts down the thermoregulation process if the pressure on the compressor side gets too high.

Applies to Unistats: Once the Unistat is ready for operation again, the high-pressure cut-out must be pressed to continue the thermoregulation process.

Applies to Unichillers: Once the Unichiller is ready for operation again, the high-pressure cut-out is automatically reset to continue the thermoregulation process. (The high-pressure cut-out in Unichillers is not readily accessible.)

High-Temperature Cooler (with Unistats):

A high-temperature cooler is a heat exchanger that cools a thermofluid from a high temperature to ambient temperature using air or water. It is installed on the upstream side of chillers in Unistats, thus reducing the load on and power consumption of the chiller.

High Temperature Stage (HT) (with Unistats):

The *high temperature stage* is the top temperature stage in multi-stage chillers in Unistats. It is used for decreasing the temperature (to approx. $-60\text{ }^{\circ}\text{C}$) and thus prepares the *low temperature stage* (NT) in dual-stage chillers (to approx. $-90\text{ }^{\circ}\text{C}$) and the medium temperature stage (MT) (to approx. $-90\text{ }^{\circ}\text{C}$) in triple-stage chillers.

The *medium temperature stage*, in turn, prepares the low temperature stage (currently to approx. $-120\text{ }^{\circ}\text{C}$) in triple-stage chillers.

Housing Volume

The volume resulting from the outer dimensions of the thermostat.

Immersion Cooler (TC)

A cooling unit with a flexible tube and a cooling coil (evaporator) for immersing in baths.

Immersion Thermostat (Polystats cc)

This is a thermostat that is combined with a bath tank that forms an independent unit. Immersion thermostats are equipped with a screw clamp for fastening to the walls of any desired bath tank. Using a bath bridge, they can also be mounted permanently on the top of a bath or on a tripod.

Industrial Thermostats (IC-Hx)

Industrial thermostats are cooling circulators (intelligent chillers) with a heater installed ex works. High cooling, heating and pumping capacities and small fluid volumes provide for fast cooling and heating rates. They are ideal for thermoregulation in process engineering and in narrow temperature ranges ($-20\text{ }^{\circ}\text{C}$.. $120\text{ }^{\circ}\text{C}$).

Interface, Analog (AIF)

This interface is used for entering the temperature setpoint or for the output of the actual temperature as an analog value in the form of a signal current ($4 - 20\text{ mA}$). Refer to: 4.3 Main Menu, Analog Interface!

Interface, Digital

This interface is used for digital data exchange between connected devices via the data communication line. Primarily, the temperature setpoint and the actual temperature are transmitted. Refer to: RS232 Interface and RS485 Interface!

Refer to: 4.3 Main Menu, Digital Interface!

Low Temperature Stage (NT)

Refer to High Temperature Stage!

Medium Temperature Stage (MT)

Refer to High Temperature Stage!

Net Refrigerating Capacity

The effective capacity the cooling thermostat or chiller provides for the application. The frictional heat generated by the circulation pump and the heat entering due to insufficient insulation has already been taken into consideration. The cooling capacity data correspond to those of the net cooling capacity.

NFL

Refer to Safety Classification!



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6.5. Huber Glossary

Definitions and Cross-References

Offset Calibration

Single-point correction of a temperature sensor at a specific temperature.

Operating Temperature Range

This is the temperature range that is limited by the permissible minimum and maximum operating temperature.

The operating temperature is the temperature to which the thermostat is permitted to heat/cool the respective thermofluid.

Overheat Point

Refer to Overtemperature Protection!

Overheating

The positive difference between the condensation temperature and the topical gas temperature of the thermofluid in the refrigeration process. (Does not refer to overheating of the thermostat!)

Overtemperature Protection

Each thermostat with a heater is equipped with overtemperature protection. The overtemperature protection operates completely independently of the controller, i.e. it cannot be influenced through the controller. If at least one temperature sensor reports that a temperature limit has been exceeded, the thermoregulation process is stopped at once.

In the course of the thermoregulation process, the sensors are tested for short-circuit and failure. The overtemperature protection is equipped with an independent protective device. This prevents very reliably that the OK status is set while there actually is a failure. The user sets the upper temperature limit (overheating point) at the thermostat to ensure operating safety considering the respective thermofluid used. For unattended operation, the overheating point must be selected at least at 25 °C below the flash point of the respective thermofluid.

Refer to: 4.1.!

Physical Mass

The physical mass describes the P/I parameters (P = proportional, I = integral) of the thermostat. Refer to 4.3!

PLC (Programmable Logic Controller)

A device frequently used in industrial environments to control operations and processes.

POKO Potential-Free Contact

The POKO is a change-over contact led through the POKO relay in the controller. The potential-free contact is designed for an ohm resistive load up to 30 V and a maximum of 0.1 A. The following applies due to the working current principle: The OK status is linked to the flow of current through the winding of the POKO relay. The POKO relay is actuated by the controller.

Presettings ex Works, Factory Default

Refer to Factory Default!

Process Control (often referred to as cascade control)

Thermoregulation of a connected application. A temperature sensor (Pt100) that forms part of this application is connected with the thermostat. The actual temperature of the external process is sensed and the operating temperature of the thermostat is permanently calculated and adapted. Depending on the operating temperature, losses through insulation and exothermic reactions, the operating temperature at the consumer may deviate considerably from the setpoint and the actual temperature of the application.

Caution!

Observe the safety-relevant limits of the respective thermofluid!

Process Temperature (core temperature)

The process temperature is the temperature measured in the core of a connected external application (in the case of external thermoregulation), e.g. in the core of a reactor. Refer to Reactor, Control Mode! Refer to 2.3!

Program (Thermostat Control)

The controller provides programs for the user to operate the thermostat, to monitor and control statuses and processes, i.e. to perform the thermoregulation process.

Program (Thermoregulation)

Refer to Thermoregulation Program!



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6.5. Huber Glossary

Definitions and Cross-References

Protocols

Protocols are used in digital data exchange. The term protocol in this context refers to a series of rules drawn up to facilitate uniform coding and decoding of the bits and bytes in the data exchange between the thermostats and a **PC**, **PCS** (process control system) or the like. For the time being, three protocols are available, two of which have been implemented in all the product families.

PPP – point-to-point protocol

LAI protocol – for communication on the bus
The third protocol is currently only available for UNISTATS on demand:

Modbus protocol (Modbus RTU by Gould)

Ramp

A desired change in temperature within a given time period. A ramp is a simple and frequently used programming of a temperature pattern.

A ramp is defined through the increase in temperature and the thermoregulation period. Once the ramp has been entered, the thermoregulation process thus programmed will start automatically and does not require any accompanying inputs.

The ramp can be defined through different combinations and sequences of the setpoint, time period and slope parameters.

Where: $\text{Time period} = \text{slope} \times \text{setpoint}$

Refer to 4.3!

Reactor

A cylindrical double-wall tank permitting the flow of thermofluid through the space between the two walls (jacket). The temperature of the thermofluid is transferred to the hollow space inside (core) via the inner wall. The user fills his/her reagents into this core with separate inlets and outlets. The reagents are then indirectly thermoregulated via the inside reactor wall.

Refrigerant

Refrigerant is used in the circuit of the refrigerating unit. It abstracts heat from the thermofluid when the compressed gas expands and evaporates in the evaporator. Since 1992 HUBER has been using exclusively refrigerants free of CFC and since 1994 even refrigerants free of HCFC (e.g. R22) that do not harm the ozone layer, i.e. with an ODP (Ozone Depletion Potential) = 0 and minimized GWP (Global Warming Potential, e.g. greenhouse effect).

Refrigerating Capacity

Heat flow led off from the thermofluid by a cooling thermostat.

Remote Control Panel

An external device permitting operation of the Huber Thermostat (possibly to a limited extent).

RS232 Interface

Digital controller interface for digital data exchange between connected devices via the data communication line. RS232 is a serial interface for establishing a point-to-point connection. This means, only two parties, e.g. a thermostat and a PC, can communicate via the interface at a time.

Refer to: 4.3. Main Menu, Digital Interface!

RS 485 Interface

Digital controller interface for digital data exchange between connected devices via the data communication line.

As many as 32 parties can be connected to the RS 485 interface. Each party connected to this bus system is assigned a separate address.

Refer to: 4.3 Main Menu, Digital Interface!

Safety Classification

In thermostats, either flammable (FL) or non-flammable (NFL) thermofluids (fluids for thermoregulation) may be used. The respective safety requirements have been laid down in DIN EN 61010-2-010. Accordingly, the safety classes NFL and FL have been defined:
NFL (non-flammable) thermostats with integrated overheating protection exclusively for non-flammable fluids.

Thermostats classified as FL (flammable) with adjustable overtemperature protection and low liquid level protection for flammable fluids (all the thermostats by HUBER). Refer to: 3.1. Thermofluid!

Shaft performance

This is the mechanical (heat) capacity introduced to the heat-transfer medium by the rotational motion of the impeller.

Segment

A segment is a section of a thermoregulation program that creates part of the temperature graph.

Refer to Thermoregulation Program!

Self Test

Test procedure in the thermostat after power on. It ensures the operability of safety-relevant thermostat components.
(Except for overtemperature protection!)



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Setpoint

The temperature to be attained at the point of measurement.

Maximum Setpoint

The highest permissible temperature to be attained at the point of measurement.

Minimum Setpoint

The lowest permissible temperature to be attained at the point of measurement.

Refer to Control Mode! Refer to 4.4!

Slave Address

Identification number the user assigns to a device to be able to allocate data to a device during data transfer via a bus protocol.

Refer to RS485 Interface!

Standards

Safety requirements for electrical laboratory equipment, and especially for thermostats, have been laid down in European Standards EN 61010-1 and EN 61010-2-010. These standards replace DIN 12879 (among others). The terms and characteristic data are defined in DIN 12876-1 and DIN 12876-2.

Status Display

Display on the UniCop that provides information on the topical program sequence or the topical status of the thermostat.

Temperature – internal, external

Refer to Control Mode, Flow Temperature!

Temperature Stability

This term denotes half the difference in temperature between the highest and lowest temperature measured in a thermostat within 30 minutes of attaining a stable value at a specific setpoint.

This value is determined at a temperature of 70 °C (with water) for a heating thermostat and at – 10 °C (with ethanol) for a cooling thermostat.

Also refer to DIN 12876!

Temperature-Stable

Refer to Thermoregulation, Temperature-Stable!

Tempmove

Possibility of selecting the input variable for a temperature.

Thermofluid (fluid for thermoregulation)

Heat transfer fluid. Transfers energy from the thermostat to the application or vice versa. Siehe 3.1.1! Refer to 3.1!

Thermoregulation Fluid

Refer to Thermofluid!

Thermoregulation, Temperature-Stable!

Thermoregulation will proceed until the defined setpoint has been attained (if required, the desired thermoregulation period will be exceeded).

Thermoregulation, Time-Stable

Thermoregulation will proceed according to the time targets entered (independent of the actual temperature value attained).

Thermoregulation

The active manipulation of the temperature of a substance to the desired level by way of heating and/or cooling

Thermoregulation Program

Arrangement of segments to create a reproducible temperature graph.

A thermoregulation program is made up of a program header and segments.

The program header is independent of the number of active segments and the length of the program.

The program header defines the beginning and sequence of the thermoregulation program (TP) through the following parameters:

Start temperature of segment n, start ramp slope (K/min),

thermoregulation priority: temperature-stable or time stable, program end.

A segment is defined through the following parameters:

Start temperature of segment n (= end temperature of segment n-1)

Segment period n

Refer to: 4.4!

Transparent Bath

A bath thermostat with transparent walls permitting direct monitoring of the object to be thermoregulated (Polystat models A5 – A18).



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6.5. Huber Glossary

Definitions and Cross-References

Usable Depth

The term usable depth denotes the depth of fluid available in the bath thermostat for direct thermoregulation.

Venting

A special circulation regime with the objective of driving out air trapped in closed thermofluid circuits.

Viscosity

Also referred to as „internal friction“. The characteristic of a fluid or gaseous substance that causes frictional tension at deformation in addition to the thermodynamic pressure, which counteracts the displacement of fluid or gaseous particles relative to each other.

This means, the viscosity of a substance increases with decreasing temperature, the available cooling capacity decreases with decreasing temperature.

The viscosity of thermofluids (fluids for thermoregulation) used in HUBER thermostats should not exceed 50 mm²/s.

Watchdog

A safety device in the electronic control, similar to the principle of the dead-man switch in an engine. The functionality of the system is monitored through the input signals.

Warning

Device messages pointing to irregularities in the operation of the thermostat that do not cause a shut-down of the thermoregulation process.
Programs 10, 11

Working Temperature Range

The working temperature range is defined with reference to an ambient temperature of 20 °C. It denotes the temperature range the thermostat attains by itself without the aid of any auxiliary means*, utilizing electric power only.

Due to the heat transferred by the pump motor and the insulation, the working temperature range of heating thermostats starts above the ambient temperature and ends at the upper limit of the operating temperature.

In the case of cooling/heating thermostats, the working temperature ranges from the lower limit of the operating temperature to the upper limit of the operating temperature. This temperature is permissible in continuous operation with the use of a chiller.

The working temperature range of cooling thermostats starts at the lower limit of the operating temperature and ends at ambient temperature. *Auxiliary means are additional heating elements, coolers, heat exchangers or fans.

Bad- und Umwälzthermostat mit luftgekühlter Kältemaschine. Eine leistungsstarke, drehzahlgeregelte Druck- und Saugpumpe ermöglicht den universellen Einsatz. Übertemperaturschutz und Niveauschutz, Sicherheitsklasse FL, Heizleistungsautomatik, Kompressorautomatik und Kälteleistungsanpassung.

Plug & Play Technologie – mit mikroprozessorgesteuertem Compatible Control Regler der neuen Generation. Einfache Bedienung mit Drehgeber und Digitalanzeige. Alarmer und Warnungen werden visuell und akustisch unterstützt. Alle Regler sind rückwärts kompatibel, über Kreuz tauschbar und kalibrierfähig. 3 Jahre Garantie. Neue Funktionen: Komfortablere Bedienung durch grafische 4-zeilige LCD-Anzeige und leistungsfähigen Prozessor. Administrator-Level und mehrere User-Level verfügbar. Diese ermöglichen sehr übersichtliche Bedienung und Schutz vor unberechtigtem Zugriff auf Parametrierung des Ministats.

Ministat230-1 ist für einfachere Temperieraufgaben konzipiert. Preisbewusst und funktionell auf das wesentliche beschränkt.

Ministat230-2; wie **Ministat230-1** jedoch mit Externfühleranschluss, Externregelung und Temperaturprogrammgeber (für 1 Programm mit max. 5 Segmenten).

Ministat230-3; wie **Ministat230-2** jedoch mit dem kompletten Funktionsumfang der Compatible Control Thermostate:

Programmgeber (50 Segmente aufteilbar auf 10 Programme), dialogfähig, verfügt über digitale Schnittstellen RS232, RS485 und analog für bidirektionale Kommunikation, sowie über potenzialfreien Meldekontakt. Lässt sich gut in Automatisierungssysteme einbinden. Geeignet für Anschluss der Huber Fernbedienung.

Bath and circulation thermostat with an air cooled refrigeration system. A powerful, variable speed, pressure and suction pump allows universal application. Over temperature protection and level protection, safety class FL, Automatic heating power regulation, automatic compressor switching and cooling power adjustment. Plug and Play technology – with the new generation of microprocessor controlled CC controllers. Simple operation with a rotary knob and digital display. Alarms and warnings are given both visually and acoustically. All controllers are backwards compatible, interchangeable and can be calibrated. 3 year guarantee. New functions: simpler user guidance using a 4 line graphical LCD display and high performance processor. Administrator level and multiple user levels are available. This enables very clear user guidance and protection from unauthorised access to the ministat parameters.

Ministat230-1 is designed for the simpler temperature control tasks. Inexpensive and limited to the essential functions.

Ministat230-2 is like the **Ministat230-1** with however an external temperature sensor connection, external temperature control and a temperature programmer (for 1 program with a maximum of 5 segments).

Ministat230-3; like the **Ministat230-2** including however the complete functions of the Compatible Control thermostat: Programmer (50 segments, may be split into 10 programs), interactive, contains a digital RS232 or RS485 interface as well as a 4 to 20mA analogue interface for bi-directional communication, and a potential free alarm contact. This allows it to be easily integrated into laboratory automation systems. Suitable for connection to the Huber remote control.

Technische Daten	Technical Data	Ministat 230-1	230-2	230-3
Arbeitstemperaturbereich	Operating temperature range		-33...200 °C	
Temperaturkonstanz bei -10 °C	Temperature stability at -10 °C		0,02 K (DIN 58966)	
Temperatureinstellung	Temperature adjustment		digital	
Temperaturanzeige	Temperature indication		digital	
Auflösung der Anzeige	Resolution		0.1 / 0.01 K	
Temperaturfühler	Temperature sensor		Pt100	Pt100
Anschluss externer Fühler	connection external sensor	---	Pt100	Pt100
Analoges Interface In/Out	Analogue interface in/out	---	4-20 mA alternativ 1-5V	
Sicherheitsklasse	Safety classification		FL	
Heizleistung	Heating capacity		2,0 kW	
Förderleistung der Druckpumpe max.	Force pump max.		18 l/min	
Förderhöhe (Druck) max.	Pressure of pump max.		0,6 bar	
Förderleistung der Saugpumpe max.	Suction pump max.		12 l/min	
Förderhöhe (Saug) max.	Suction max.		0,3 bar	
Pumpenanschluss	Pump connection		M16x1	
Pumpenanschluss für Schlauch	Pump connections for hose		NW8/ NW12	
max. zulässige kin. Viskosität	max. permissible kin. viscosity		50 mm ² /s	
Badvolumen	Bath volume		3,4 l	
Badvolumen mit Verdränger Einsatz	Bath volume with displacement rack		1,9 l	
Badöffnung BxT	Width bath opening WxD		170 x 83 mm	
Badtiefe	Bath depth		135 mm	
Abmessungen B x T x H	Overall dimension w x l x h		255 x 450 x 400 mm	
Netzanschluss	Power supply requirement		220-240 V 1~ 50/60 Hz	
max. Stromaufnahme	max. current (1 phase)		11,8 A	
Absicherung	fuse		16 A	
Schutzklasse	Protection class		IP20	
max./min. Umgebungstemperatur	max./min. ambient temperature		30 °C / 5 °C	
Gewicht	weight		35kg	
Bestell-Nr.	Order-No.	741.0001	741.0002	741.0003
				V1.0/03

Zubehör* und Peripherie: Schlauchstutzen NW8*/ NW12*, Blindstopfen, Überwurfmutter M16x1*, Verbindungsschläuche, externe Fühler (nicht bei cc1), Verbindungskabel, Absperrhahn, Verdrängereinsatz zur Reduzierung des Badvolumens, Kalibriereinsatz *im Lieferumfang enthalten

Accessories and periphery: Adapter nom. dia 8 mm*/ 12 mm*, dummy plugs, sleeve nuts thread M16x1*, connection tubes, external sensor (not with cc1), connection cable, stopcock, displacement rack for reduction of the bath volume, calibration insert *standard equipment

Leistungsangaben gelten bei: Wassereintritt 15 °C 3bar, Umgebungstemperatur 20 °C

Ersatzteilliste / List of spare parts

ab Fert.Nr./from serial No. 55635

ministat230

Pos.	St.	Ident.Nr.	Bezeichnung	Description
1		3009	Kompressor	compressor
2		1922	Lüfter	fan
1		0421	Verschlussschraube	screw plug
1		0422	Viton O-Ring Ø10x2,5	o-ring Ø10x2,5
2		6088	Blindstopfen	blank plugs
2		6089	Überwurfmutter M16x1	sleeve nut M16x1
1		45110	Verflüssiger komplett	condenser complete
1		13599	Magnetventil	solenoid valve
1		13618	Magnetspule	solenoid coil
1		13673	Pumpenmotor	pump motor
1		13828	Heizung	heater
1		13671	Primärgetaktetes Schaltnetzteil	primary switching power supply
1		0957	Entstörglied	interference suppressor
1		0448	Anlassrelais	starting relay
1		0449	Anlasskondensator	starting condenser
1		15049	Fühler Pt100 komplett Bad	sensor Pt100 complete bath
1		15053	Fühler PT1000 kpl. Übertemperaturschutz	sensor overtemperature protection PT1000 cpl.
1		2033	Fühler Pt100 komplett Kondensation	sensor Pt100 complete condensation

741.0001

ministat230-1

1	658.0001	Regler CC1	controller CC1
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741.0002

ministat230-2

1	658.0002	Regler CC2	controller CC2
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741.0003

ministat230-3

1	658.0003	Regler CC3	controller CC3
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